Disruptive and Transformative Role of Semiconductor and Related Manufacturing in Aspiring the goal of Atamanirbhar Viksit Bharat

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Guest Lecture: EE 225

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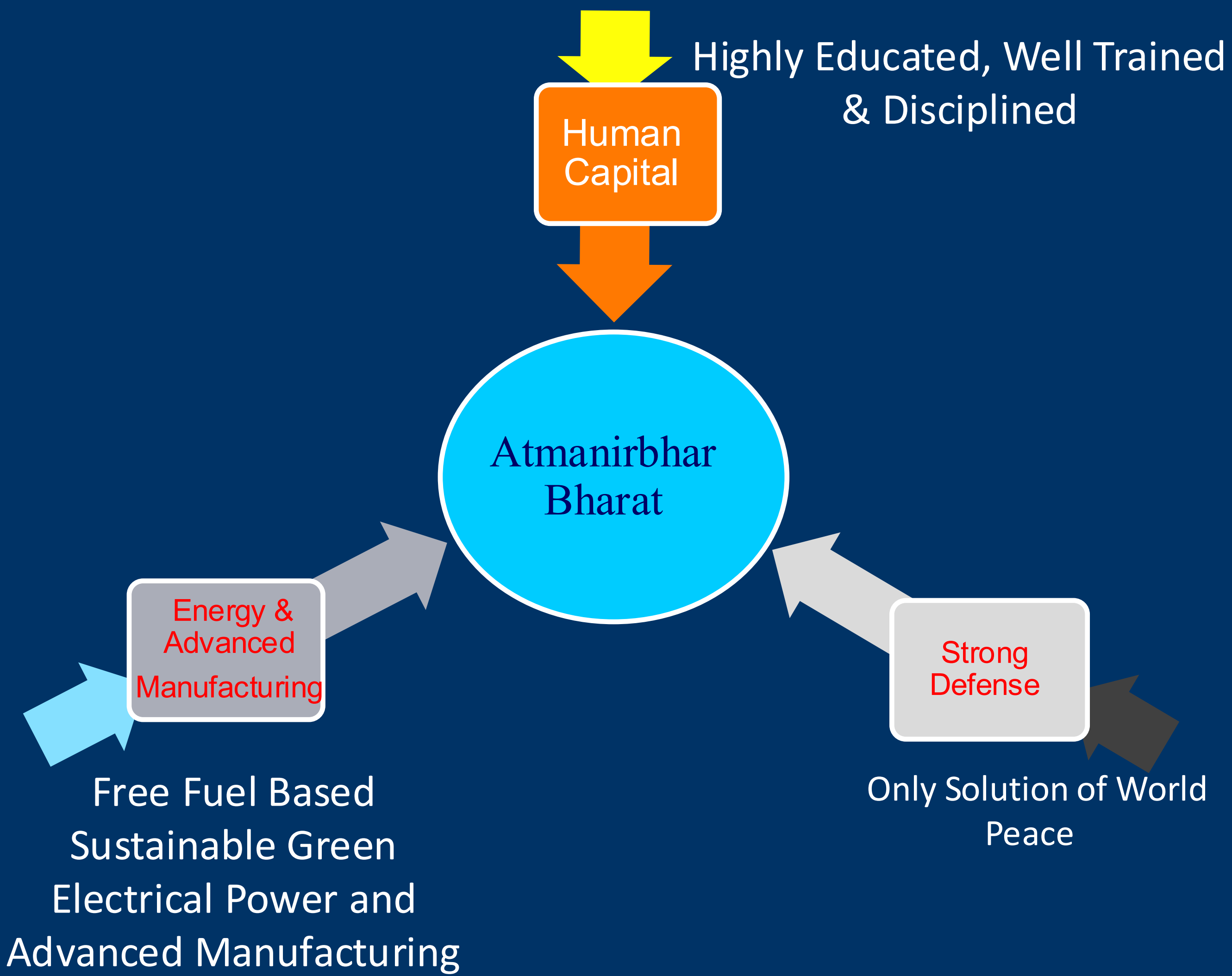
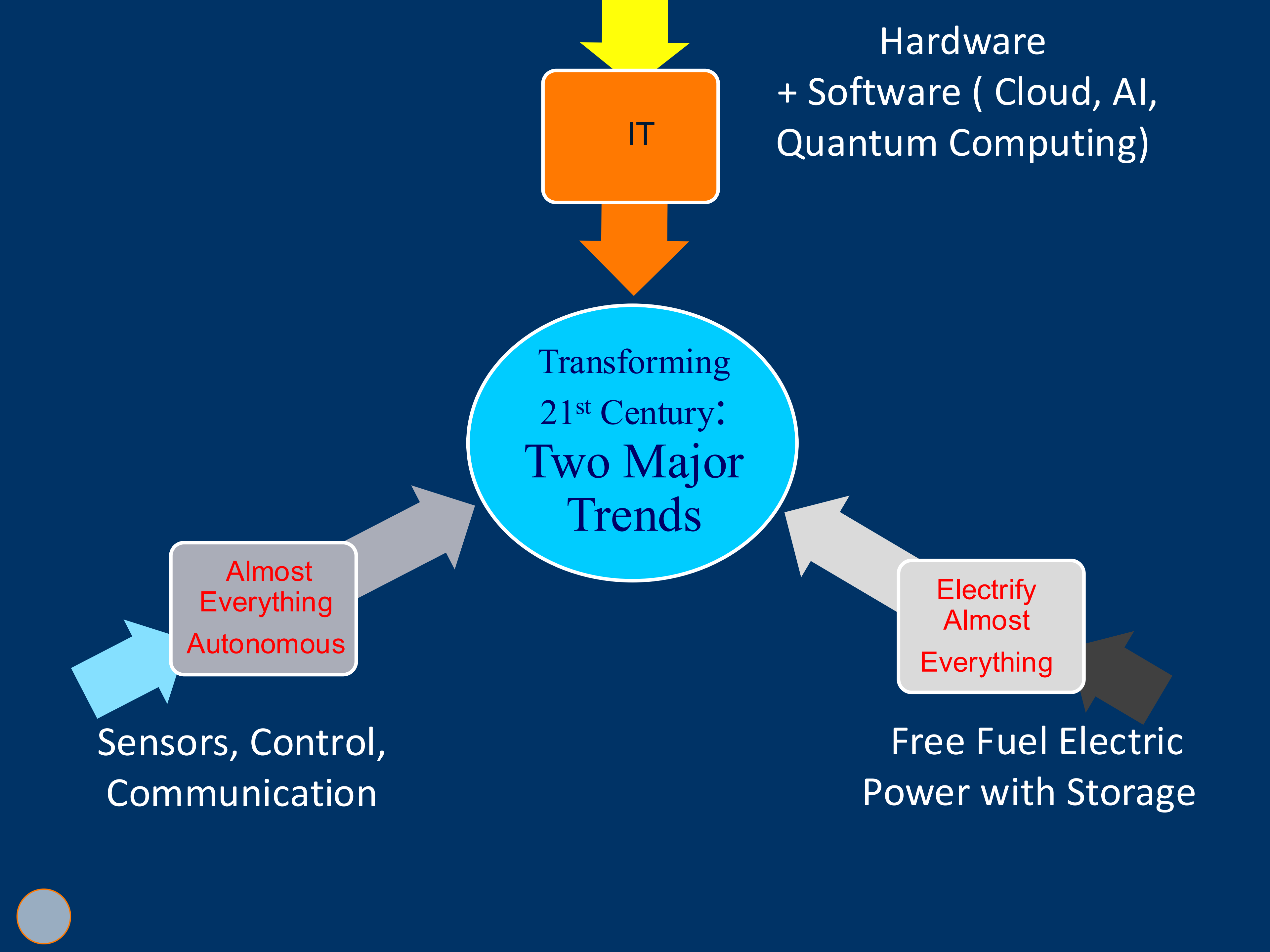
ACKNOWLEDGMENT

- A large number of Graduate and Undergraduate Students
- Former Post Doctoral Fellows & Research Associates
- Colleagues in Academia , Industry, Federal Labs and Government

Why Bharat will Become Viksit and Atamanirbhar ?

- Geopolitics of the world order is **changing** very fast: World is today Multipolar
- Bharat is a key leading partner in such a multi-layered international system
- Bharat is looking not only the welfare of its own citizens, but providing leadership in the growth of many underprivileged people living in poor countries of other continents.
- The economy of any technical solution if proven in Bharat, can be duplicated anywhere in the world. This single point will make Viksit & Atamanirbhar due to innovation.
- Bharat's economy is currently the **world's fastest-growing major economy**, with a recent GDP annual growth rate of **8.2%** for the July-September quarter of 2025
- With 1.4 Billion People, Bharat has positioned itself as the world's "Human Resource Capital ", through continued focus on quality education and development. You are part of "Human capital".
- Bharat is a global leader in digital innovation and services.
- A core principle in culture of Bharat , reflected in ancient wisdom and modern ideals, is the focus on universal welfare, summarized by concepts like "**Vasudhaiva Kutumbakam" (the world is one family) : Concept of Vedas**


Highly Educated, Well Trained & Disciplined
Human Capital
Atmanirbhar Bharat
Energy & Advanced Manufacturing
Strong Defense
Free Fuel Based Sustainable Green Electrical Power and Advanced Manufacturing
Only Solution of World Peace


IT
Hardware
+ Software (Cloud, AI, Quantum Computing)
Transforming
21st Century:
Two Major Trends
Almost Everything Autonomous
Electrify Almost Everything
Sensors, Control, Communication
Free Fuel Electric Power with Storage

52 Years of Experience of Energy & Semiconductor Industries
(Energy & Information are two sides of the same coin)

My Own Background

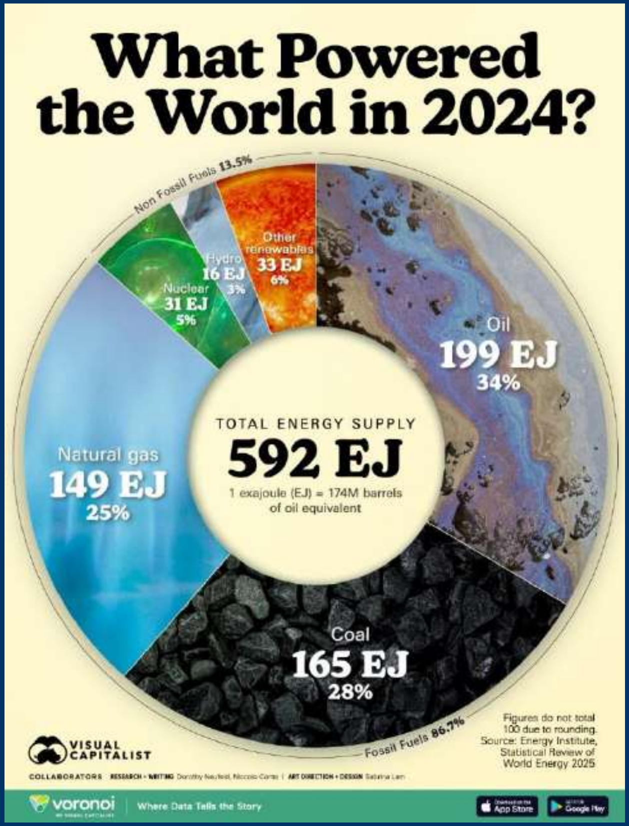
- Overseas Citizen of India. Left India in 1973
- During energy crisis of 1973 decided to do Ph .D. decertation in the area of Solar Cells. Fundamental work done by me and Prof. Green of Australia (same thesis adviser) led to TopCon solar cells. More than 60 % PV market is based on these cells.
- In 1980 I published a paper demonstrating silicon as the PV material. More than 95 % PV market is based on silicon.
- Demonstrated the importance of high-K gate dielectric to replace SiO2 as the dielectric in 80s . In 2005, Industry adopted high-k gate dielectric in CMOS manufacturing.
- Played Global Leadership Role in Developing technology of rapid thermal processing. Currently, over Billion dollar Industry.
- In 2005, Tried to establish 90 nm , 3 Billion Dollar Chip Manufacturing Plant in India. Failed due to lack of Support of the Government of India.
- In 2010 as one of Ten Global Photovoltaics Technologist (selected by PV world) predicted PV will take over wind energy that has already happen.
- In 2014 published, “Why and How Photovoltaics will provide Cheapest Electricity in 21st Century” : Utility scale electricity generation all over the world: \$0.01-\$0.2/kWh
- In 2005, with CEO of Tata Electronics as co-author published Article in IEEE Spectrum on Single wafer processing, which is lead technology of sub-5 nm CMOS manufacturing
- In 2015, As Technical Chair Founded IEEE Int. Conf. on DC Microgrid (8th, China, June 2026)
- Chair of IEEE Power & Energy Society Working Group on End-to-End DC Power Networks
- On July 13, 2025 as Chief Guest delivered convocation address of 27 the Convocation of IIT Guwahati. https://www.youtube.com/watch?v=qTaTW2vW8Jc

Selected Semiconductor Products

- Diodes
- Transistors
- Integrated circuits (computer chips) Based on Silicon
- High Speed Integrated circuits Based on GaAs, and InP
- Power Transistor Based on Silicon (IGBT)
- Power Transistor Based on Silicon Carbide
- Lasers and Other Optical Devices
- Light Emitting Diodes
- Detectors and Sensors
- Rad hard electronics for Space
- Micro-electromechanical Systems (MEMS)
- Solar Cells
- Inverters and Rectifiers
- Semiconductor Devices for Harsh Environment
- Batteries (Lithium Ion based)

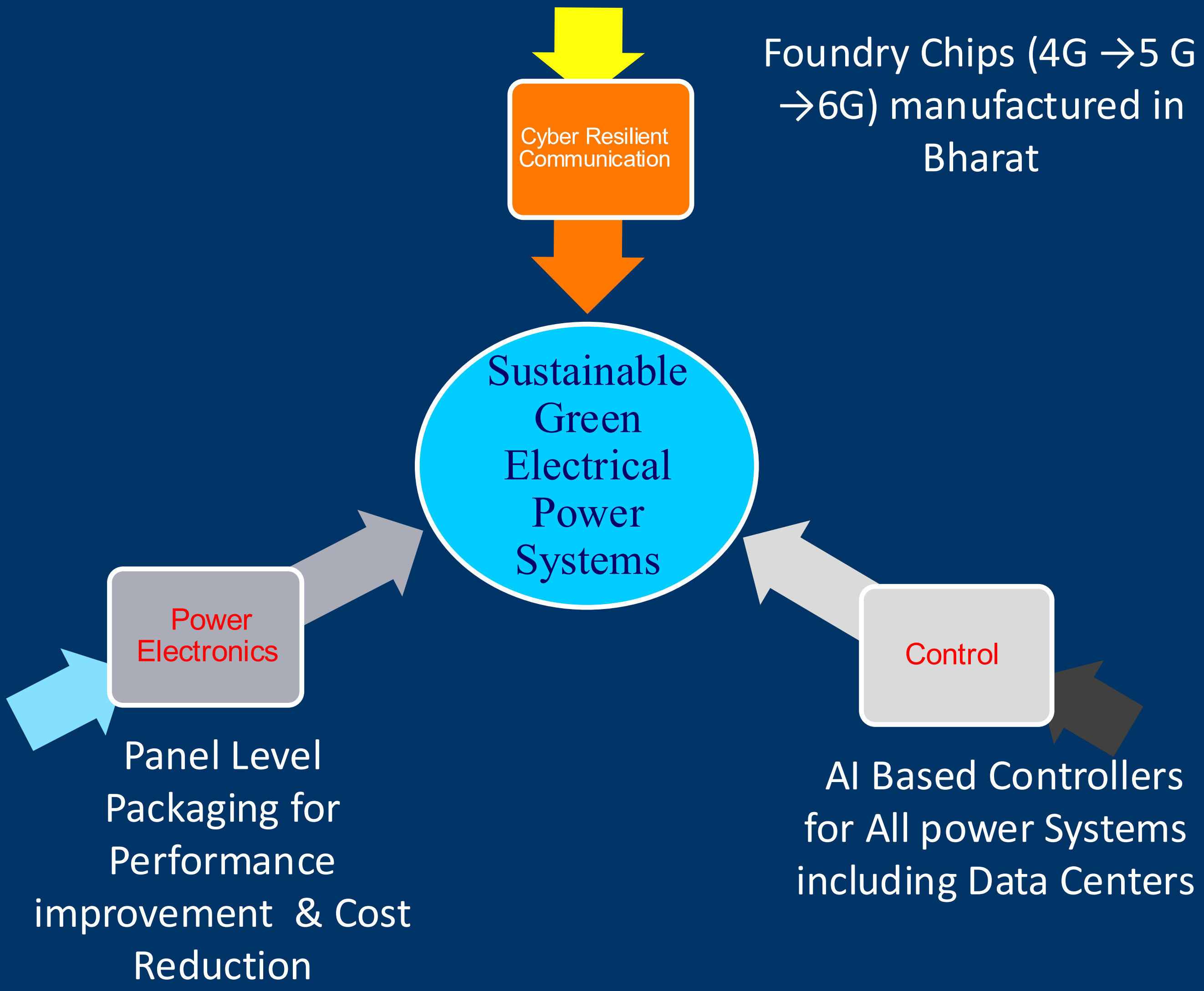
What Powered the World in 2024 ? (August 22, 2025): 86 % Dirty Energy ; India: 78 %

https://www.visualcapitalist.com/what-powered-the-world-in-2024/



Opportunity to Replace 78 % Dirty Energy by Green Electric Power in Bharat & Export Opportunities: Innovation is the key Point

- Electricity generated by Solar and wind with Storage can meet needs of everyone on planet earth.
- Current rate of transition is slow. Both public and private investment as a whole can not be increased (more than already planned).
- This means the energy efficiency of Power System must be increased
- Replacing AC power by DC power is low hanging fruit in improving energy efficiency (means Financial efficiency) and save Capital investment as well cost to consumer
- Rajendra Singh, “ Expediting Green Energy Transition”, EXPERT VIEW, IEEE Power Electronics magazine, June 2024

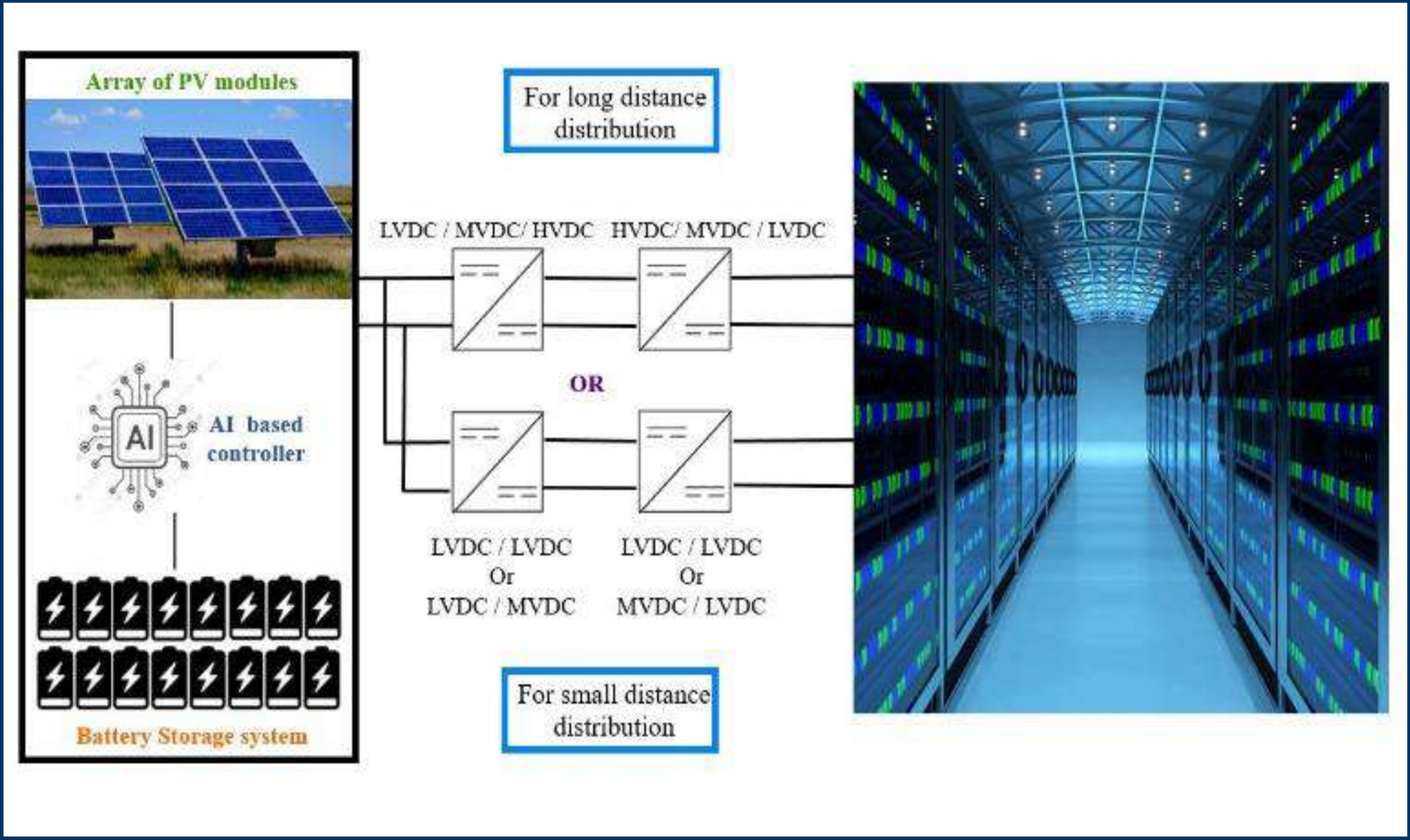


Cyber Resilient Communication
Foundry Chips (4G →5 G →6G) manufactured in Bharat
Sustainable Green Electrical Power Systems
Power Electronics
Panel Level Packaging for Performance improvement & Cost Reduction
Control
AI Based Controllers for All power Systems including Data Centers

My Work in Battery Manufacturing

- Amir A. Asif and Rajendra Singh, “Further Cost Reduction in Battery Manufacturing”, *Batteries* **2017**, *3*(2), 17; https://doi.org/10.3390/batteries3020017
- After Receiving Hind Rattan Award in 2019, I stressed the importance of Lithium Ion Batteries manufacturing in India (video Available on Request)

Local Power (PV and Battery) as Envisioned by Thomas Edison can be Implemented Today



In 2005, I Failed to Establish 90 nm Fab in India ($3 Billion Investment)

सत्यमेव जयते

अपर सचिव
संचार एवं सूचना प्रौद्योगिकी मंत्रालय
सूचना प्रौद्योगिकी विभाग
इलेक्ट्रॉनिक्स निकेतन
6, सी०जी०ओ० कॉम्पलेक्स, नई दिल्ली–११०००३
Additional Secretary
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Department of Information Technology
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Tel. : (011) 24363078, Fax : (011) 24363101
E-mail : mmn@mit.gov.in, Web. : www.mit@gov.in

3(4)/2005-IPHW
July 26, 2005

Dear Dr. Rajendra Singh,

Thank you for your letter dated 23.6. 2005 addressed to Hon'ble Minister, Communication & IT, enclosing a proposal to set up a state-of-the-art 90 nm, Semiconductor Fab in India in collaboration with IBM and Applied Materials. We are glad to know of your technology, research and intellectual property collaboration with IBM to support this venture for the next 20 years.

We have examined your proposal and the Government of India would be happy to support the proposal to ensure successful seeding of the project. Our policies are designed, and will be calibrated from time to time, to provide the necessary infrastructure for projects such as these. At a suitable time, we could discuss the terms and conditions of Government participation in the project.

I hope this commitment from the Government of India will help you in achieving the necessary financial commitments from your investors and strategic partnerships with your key technical partners.

Government of India would be happy to render all assistance to ensure that the project is implemented in the time frame envisaged in your proposal.

You had indicated that you would be able to give a time table, covering about six months, during which you would be able to take important steps towards the project. Government would be happy if you could indicate the time table in your reply to this letter.

With Regards,

Yours Sincerely,

(M.M.Nambiar)

Dr. Rajendra Singh, Ph.D., Houser Banks Professor
Director,Centre for Silcon Nanoelectronics
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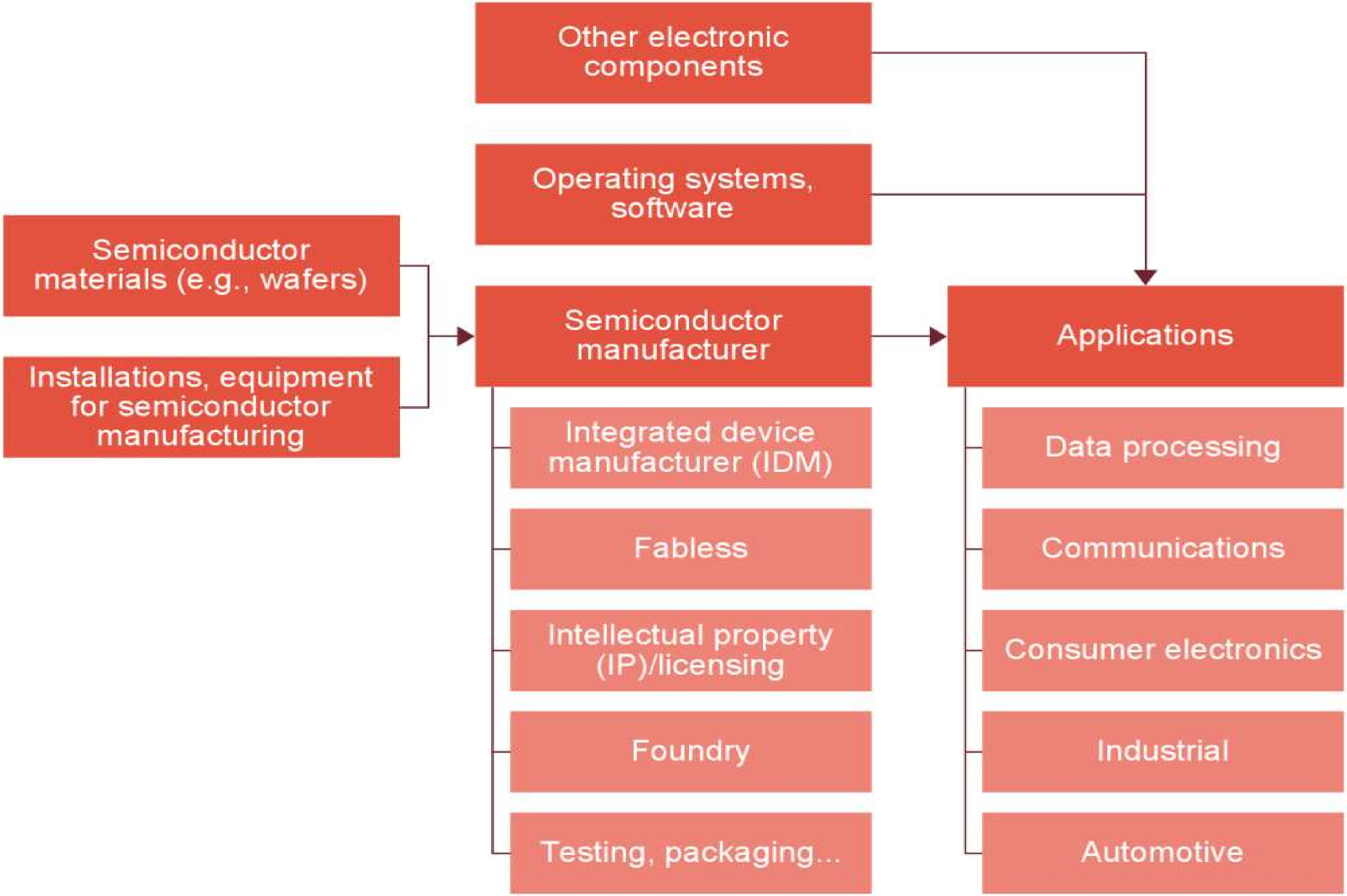
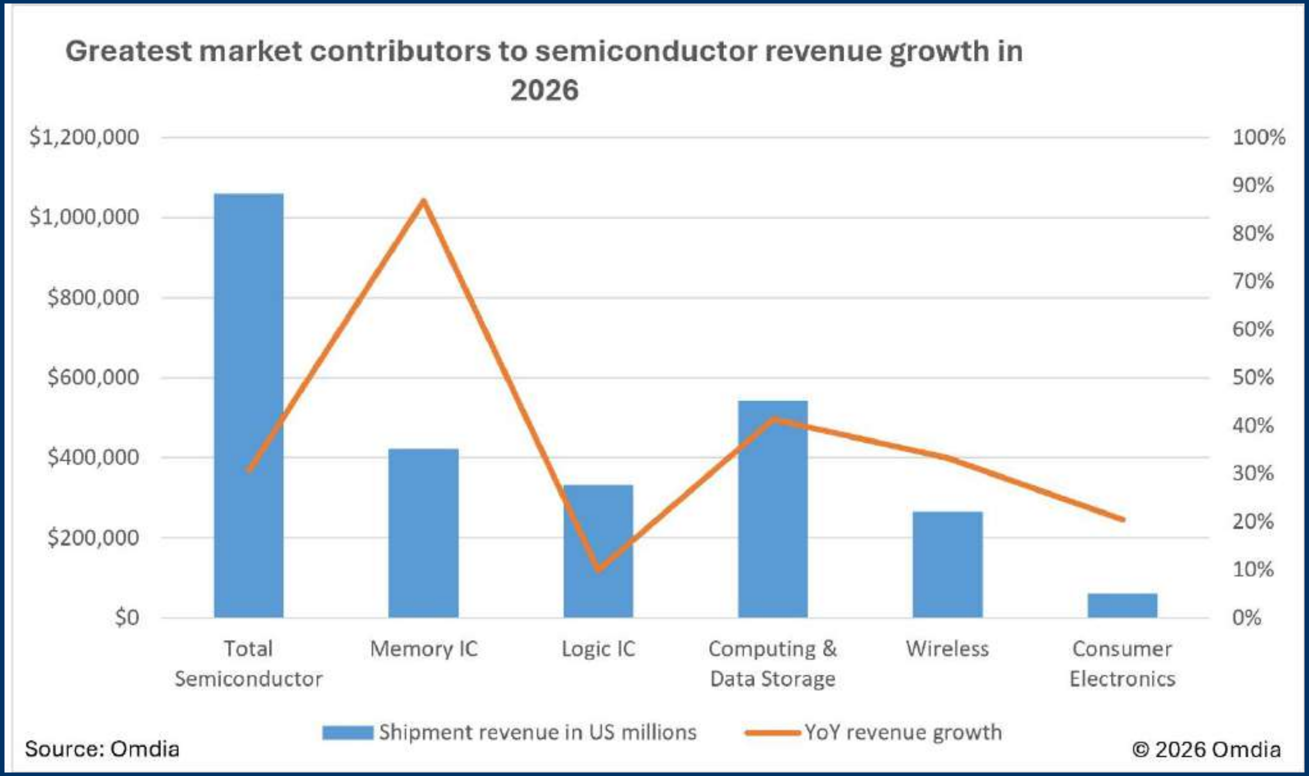
Complex Nature of Semiconductor Industry



Source: Deutsche Bank, PwC presentation

AI Drives Semiconductor Revenues Past $1 Trillion for the First Time in 2026 (Jan. 15, 2026)

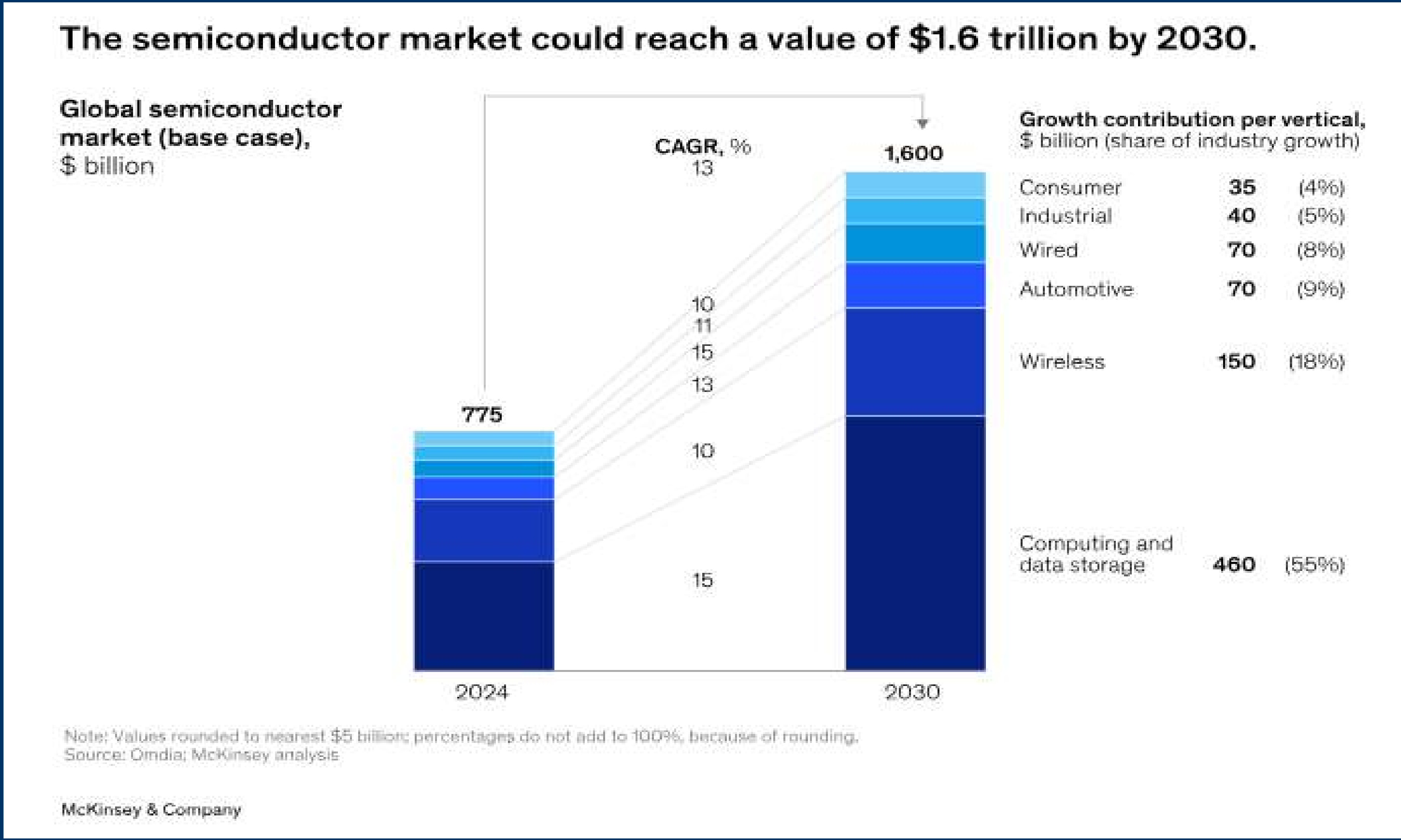
https://www.businesswire.com/news/home/20260115968050/en/AI-Drives-Semiconductor-Revenues-Past-%241-Trillion-for-the-First-Time-in-2026



A $1.6 trillion semiconductor market in 2030

January 16, 2026

https://www.mckinsey.com/industries/semiconductors/our-insights/hiding-in-plain-sight-the-underestimated-size-of-the-semiconductor-industry

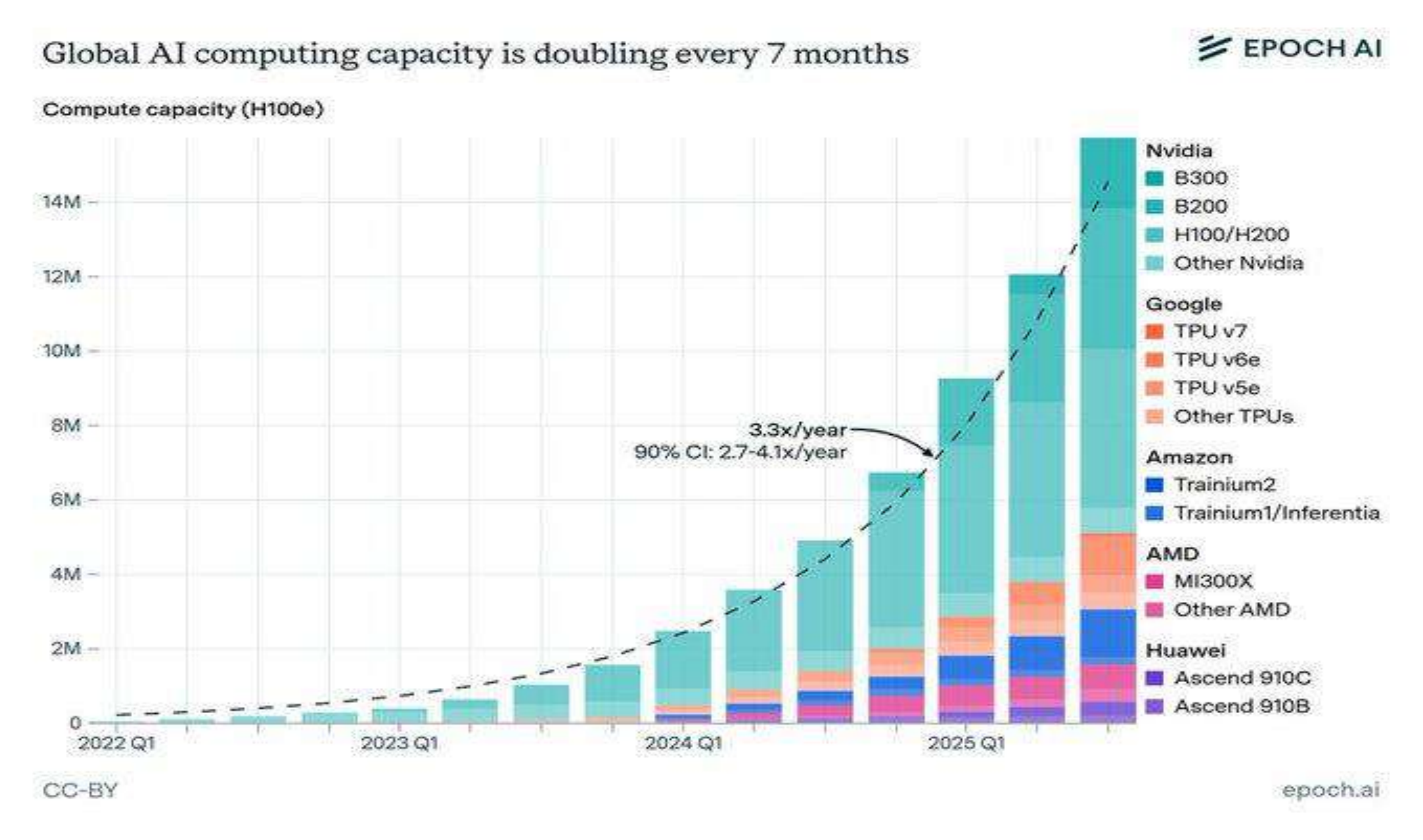


Transformative Roles

Device	Sector	Remark
Silicon (CMOS)	Information Technology	About 92- 95% of electronics worth $3.7-3.8 trillion is based on CMOS technology
Silicon PV system	Power generation	Free fuel ~ 1,000 MW (1 TW) installations mid 2022, 3 TW by the end of 2025
Lithium ion battery	Energy storage Leading to Many Transformation	~$100/kWh, nearing cheaper than Hydro Cost < $60-70/kWh New Applications

Global AI computing capacity is doubling every 7 months (January 9, 2026)

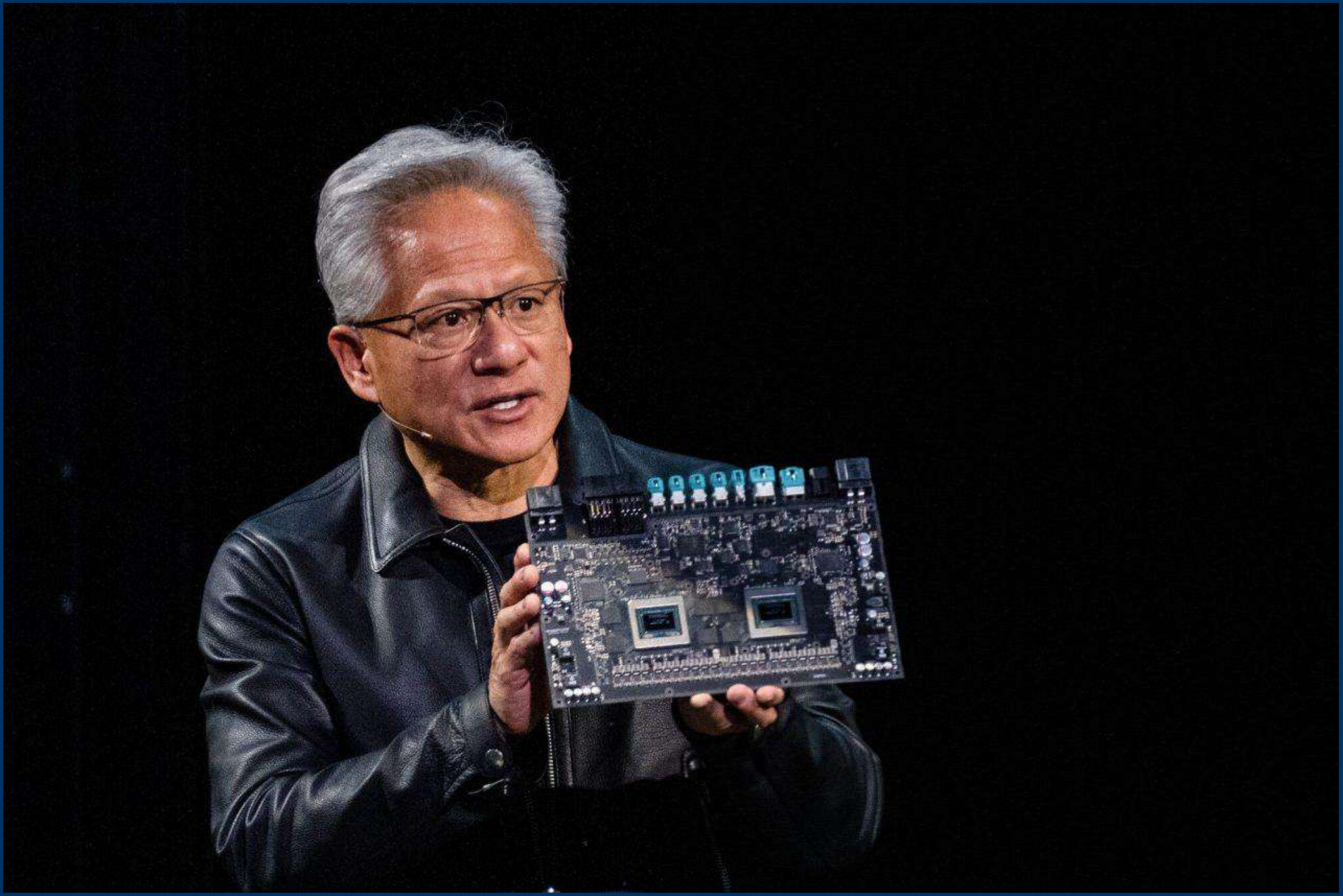
https://epoch.ai/data-insights/ai-chip-production



The Microchip Era Is About to End

The future is in wafers. Data centers will be the size of a box, not vast energy-hogging structure

WSJ, Nov. 3, 2025



Some Important Terms-1

- Three important type of materials:
- Conductor: (metal, superconductor) : Carries current, like a pipe carries Water
- Semiconductor: Controls flow of electricity , like a Faucet control flow of water
- Insulator: An insulator stops flow of electricity like a plug blocks flow of water
- *Metals*: Cu, Ag, Au etc. : For example, Resistivity of Cu; 1.8E-6

Ohm x cm

Semiconductors: Si, C, Ge, GaAs, InP, SiC etc. ;Typically !E-4 to 1E4 Ohm x cm

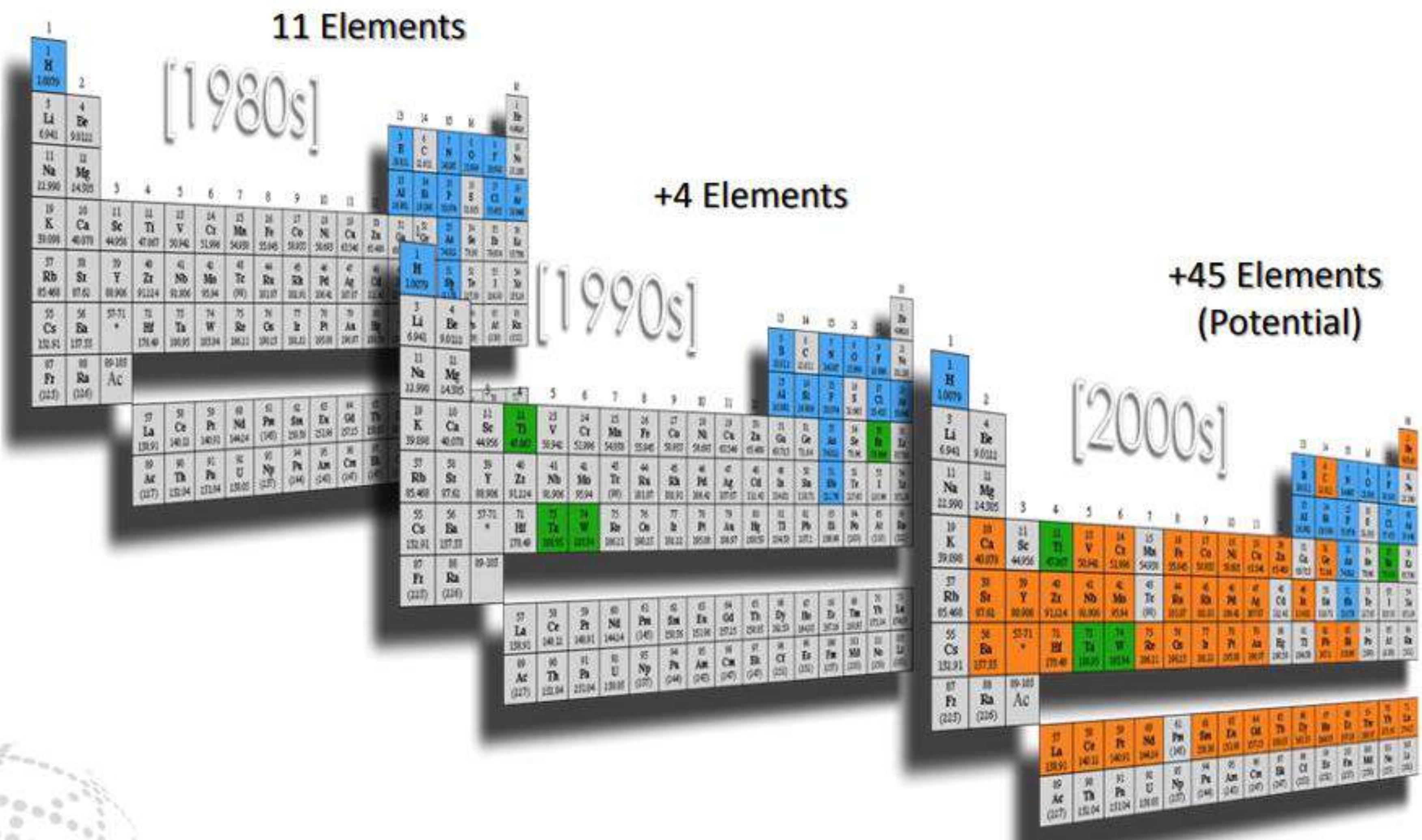
- *Insulators: Very high resistivity: Silicon Dioxide: Resistivity ; 1E-18 ohm x cm*
- Diode: Two terminal Semiconductor Device, Very small current with negative bias, Current Flows with forward bias;
- Transistor: 3 terminal Device : Think like the flow of water in a river (depends on rain and or snow melting). If you built a Dam, water flow can be controlled.
- FIELD EFFECT TRANSISTOR
- Source: Where water starts
- Drain: Where water ends.
- Gate: Control the flow of water
- The Distance between Source and drain: Chanel Length, Feature Size : Also called Technology Node

Some Important Terms-2

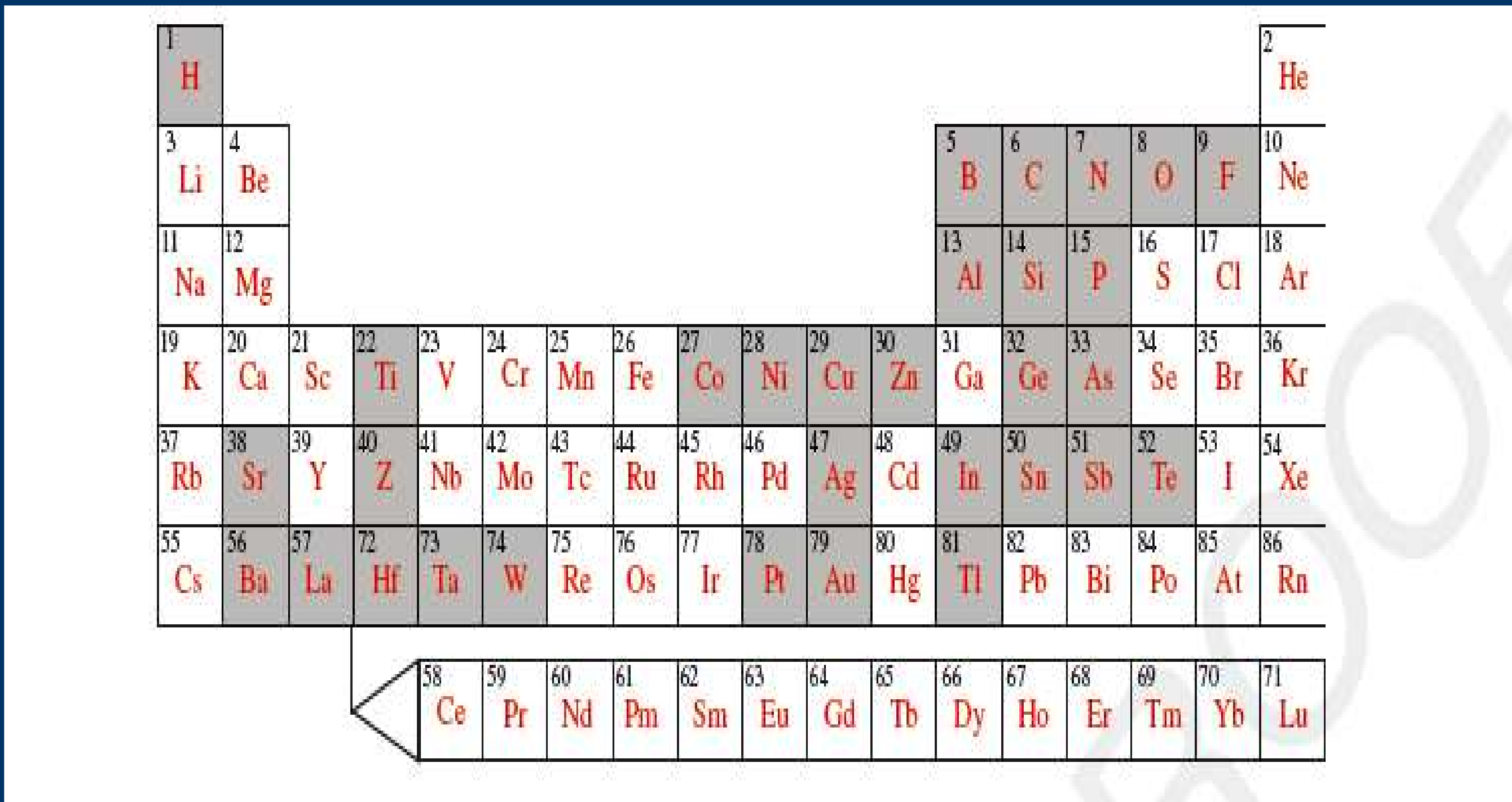
- There are other type of Transistors: we will not get into details
- The very fact that small change in height of the gate of Dam allows lot of water flow is called Gain. Very important fact ,
- At zero voltage , the current of a diode should be zero. Similarly , for transistor, for zero voltage between source and drain current should be zero .
- In reality there is current flow at zero bias , which we call leakage current. Lower vale of leakage current means device performance will improve.
- As Designer , we have constantly reduced the value of leakage current , I_L . This has allowed lower Power Supply of the Transistor (currently ~1 Volt)
- Band Gap (Eg) is important concept for semiconductor
- $I_L \propto A \cdot T2 \cdot \exp(-Eg/(2kT))$
- Higher Eg means lower leakage current
- Higher Temperature means higher value of Leakage Current

Innovations in silicon manufacturing

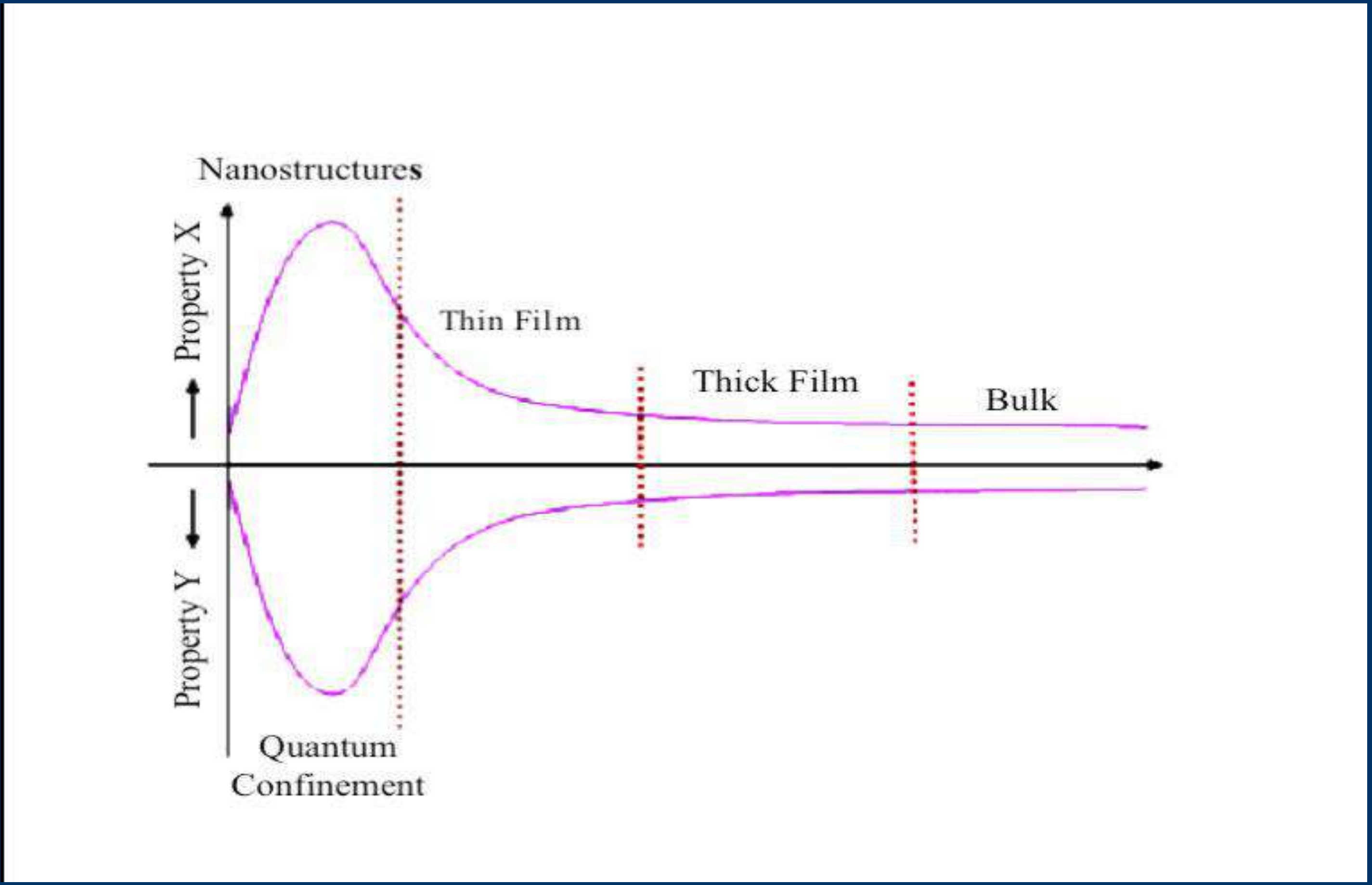
new materials enable innovation



Summary of Elements (shaded) Used in Silicon Device Technology (Guide to the State of the Art Electron Devices, p. 75, 2013)



Properties of a Material as a Function of Thickness



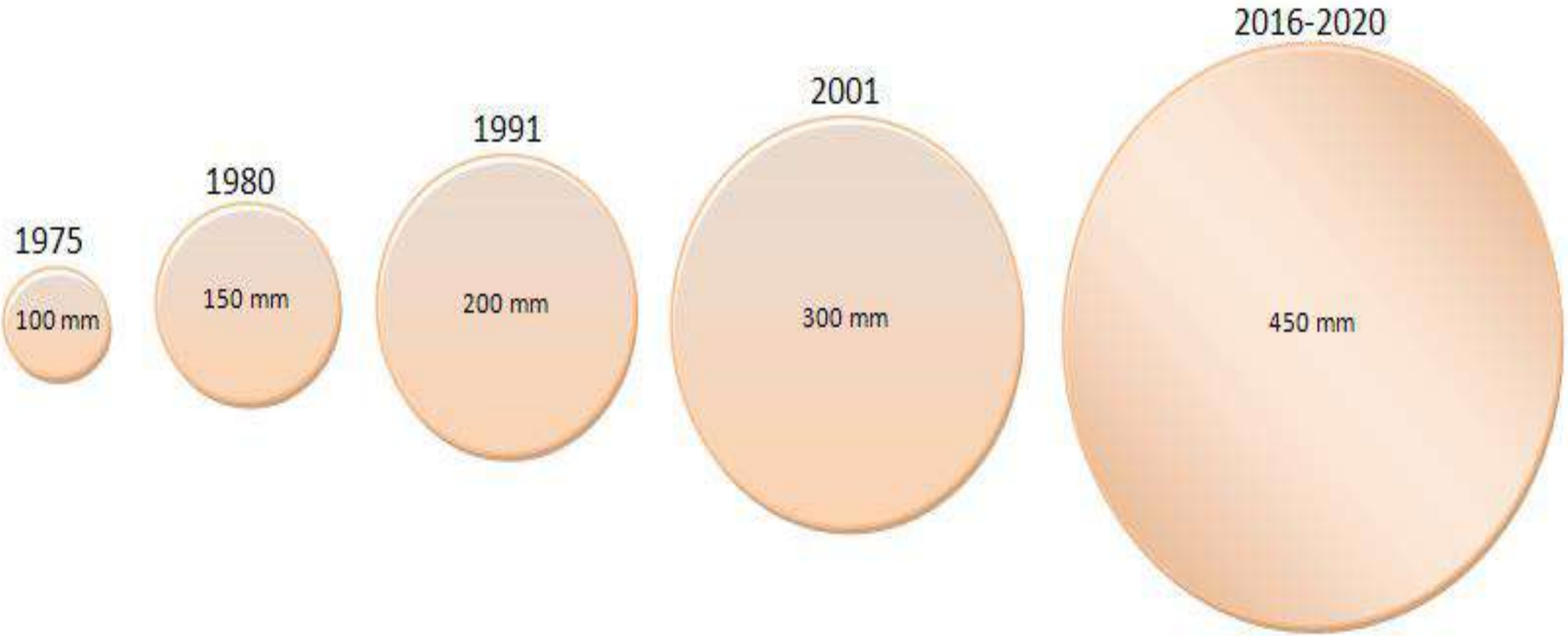
R. Singh et.al, “Dominance of silicon CMOS based semiconductor manufacturing beyond international technology roadmap and many more decades to come” Semiconductor Fabtech, 30th edition, pp. 104-113, 2006.

Some Background References

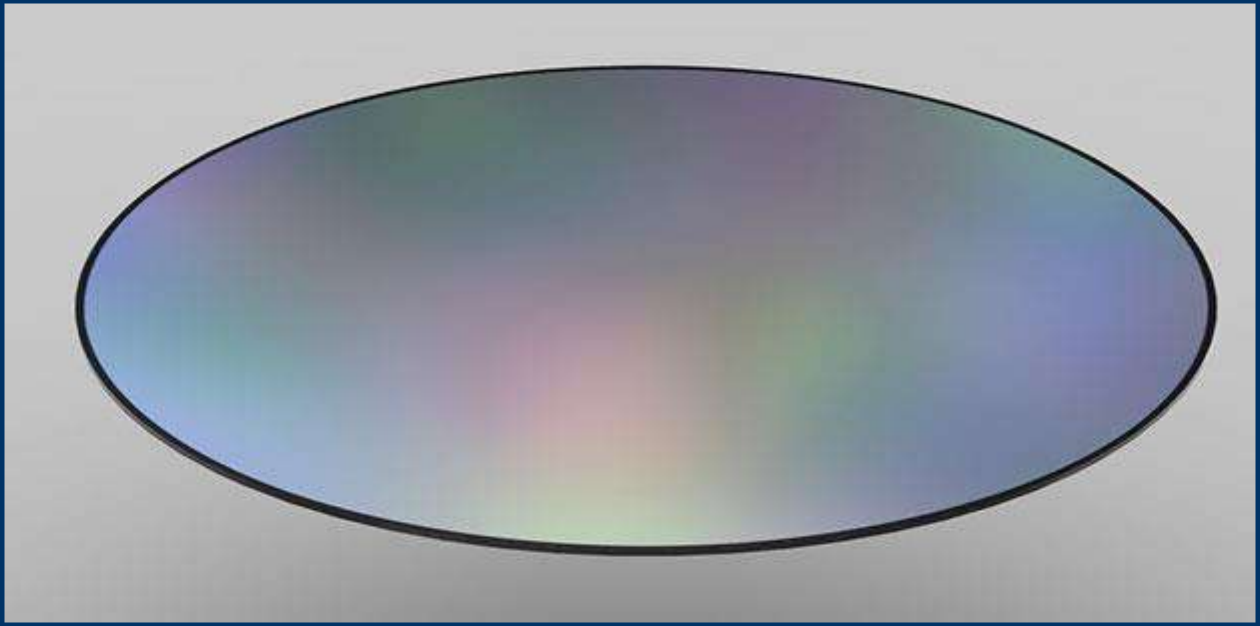
- **How Raw Silicon Wafers Are Manufactured**
- https://www.electronicdesign.com/featured-media/video/21178449/how-raw-silicon-wafers-are-manufactured

- **Basics of Semiconductor Manufacturing**
- R. Singh, L. Colombo, K. Schuegraf, R. Dowering, and A. Diebold "Semiconductor Manufacturing", Chapter 10, EDS Guide to State-of-the Art Electron Devices, pp. 121-134, Edited by J. N. Burghartz, published by Wiley, IEEE Press, 2013

450 mm Wafers not expected in near Future

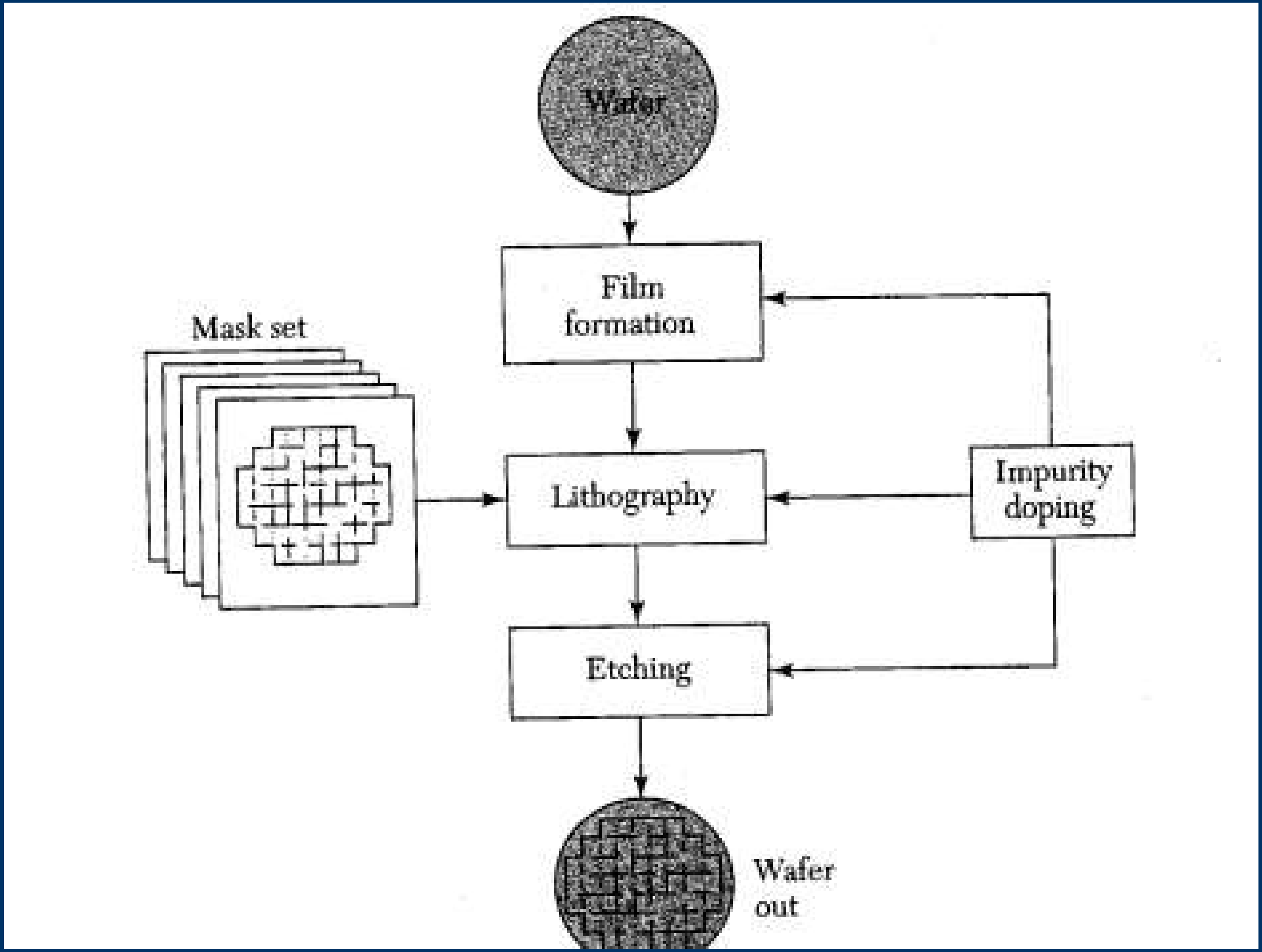


Wafer

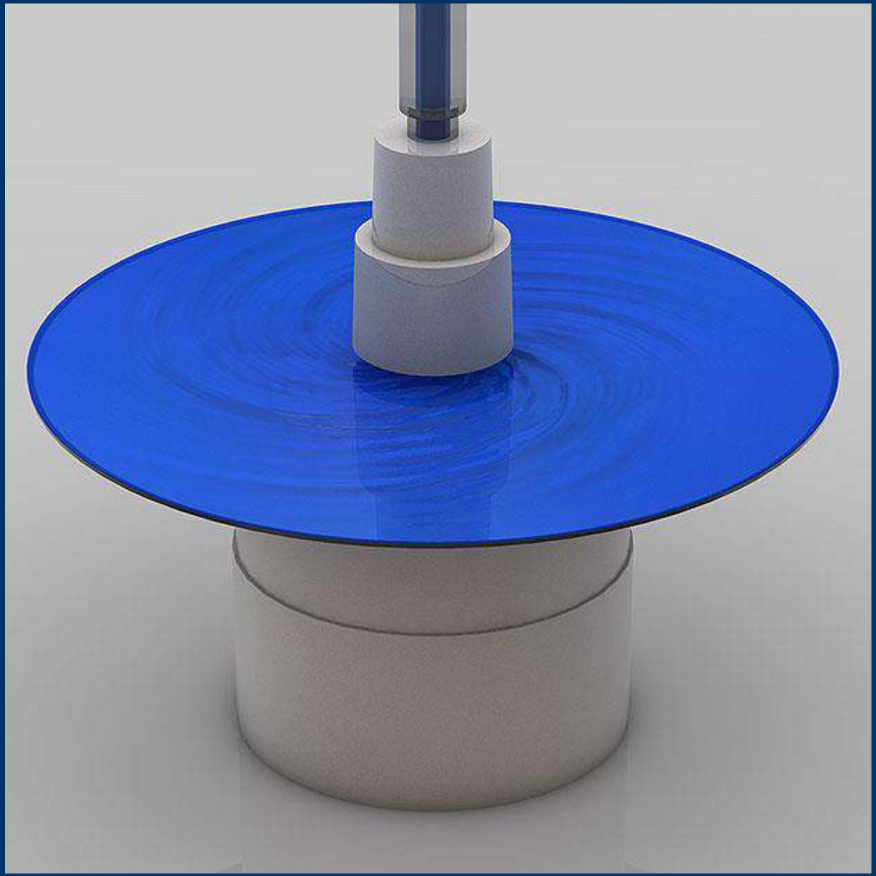


The wafers are polished until they have flawless, mirror-smooth surfaces. Today we use wafers with a diameter of 300 millimeter (~12 inches). When Intel first began making chips, the company printed circuits on 2-inch (50mm) wafers. Now the company uses 300mm wafers, resulting in decreased costs per chip.

Schematic Flow Diagram of Integrated Circuit Fabrication

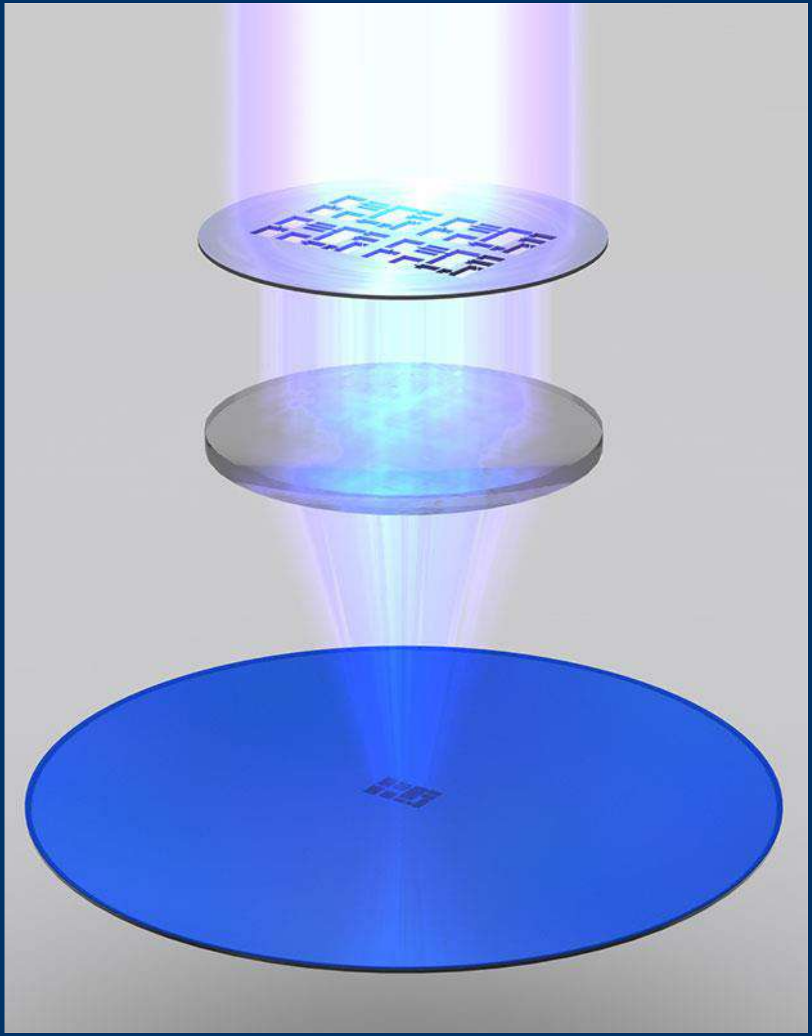


Applying Photo Resist



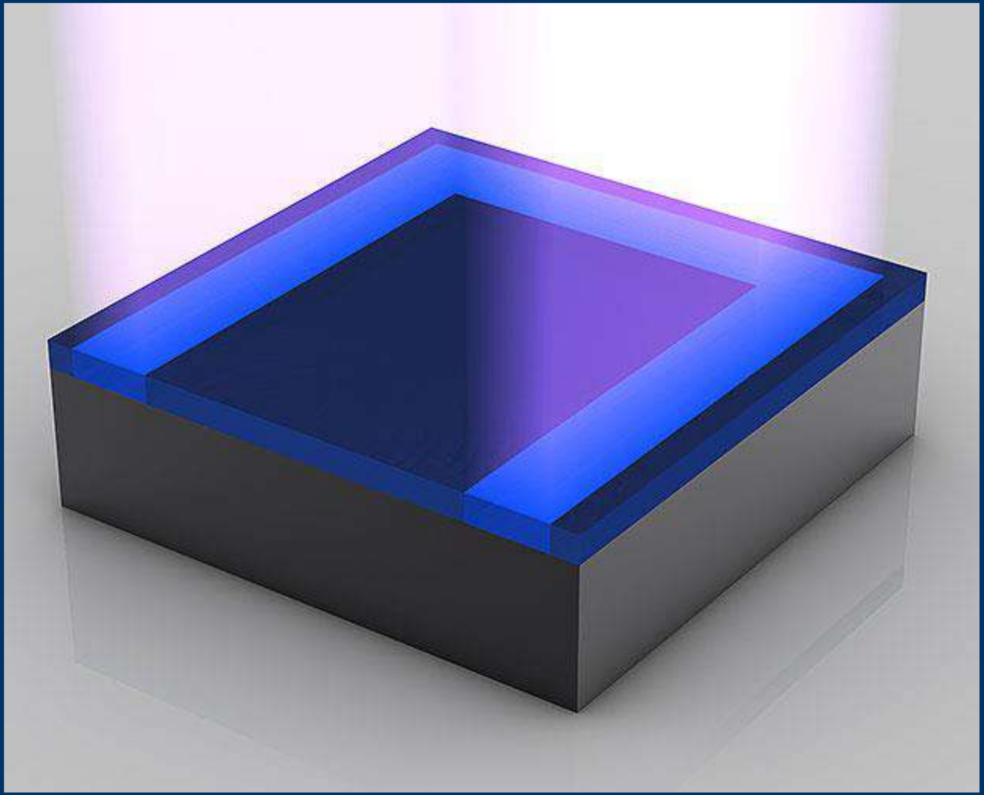
The liquid (blue here) that’s poured onto the wafer while it spins is a photo resist finish similar as the one known from film photography. The wafer spins during this step to allow very thin and even application of this photo resist layer.

Exposure



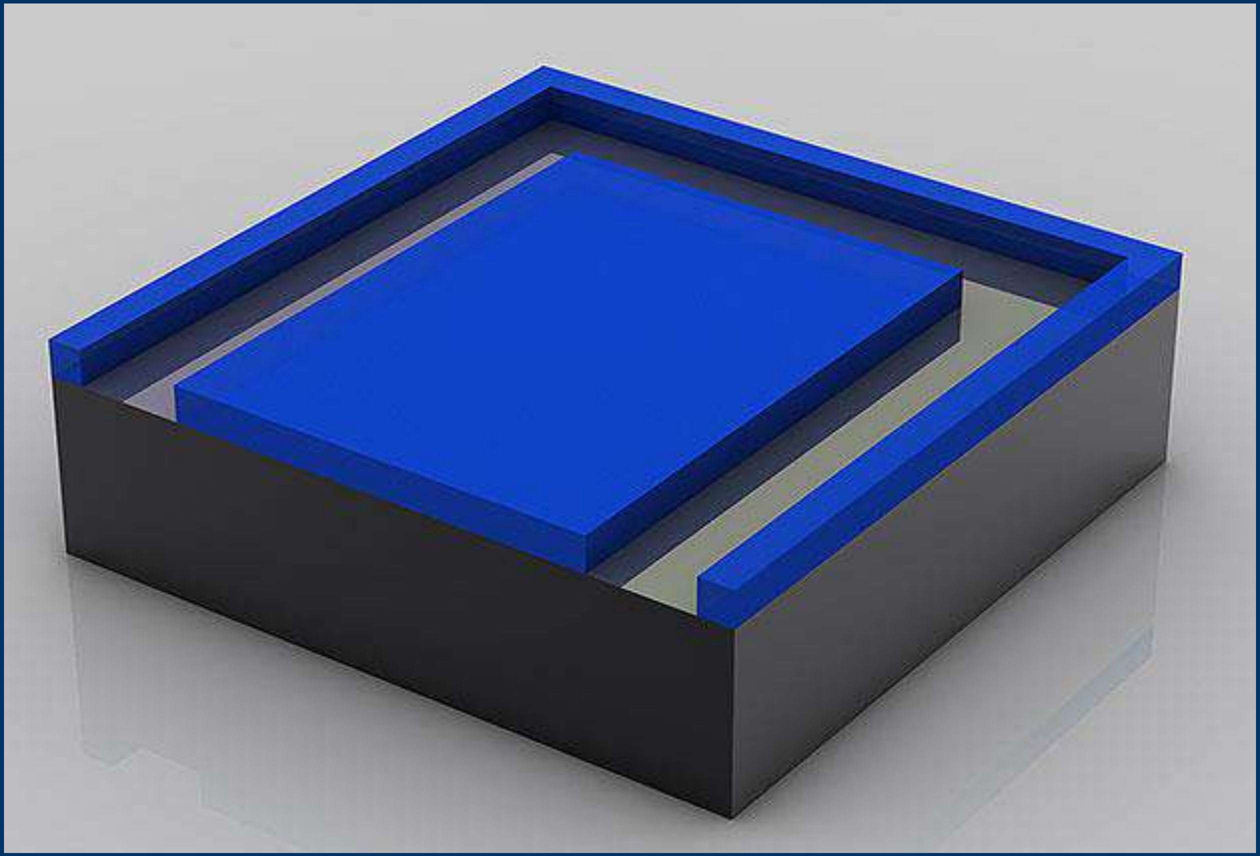
The photo resist finish is exposed to ultra violet (UV) light. The chemical reaction triggered by that process step is similar to what happens to film material in a film camera the moment you press the shutter button. The photo resist finish that's exposed to UV light will become soluble. The exposure is done using masks that act like stencils in this process step. When used with UV light, masks create the various circuit patterns on each layer of the microprocessor. A lens (middle) reduces the mask's image. So what gets printed on the wafer is typically four times smaller linearly than the mask's pattern.

Exposure



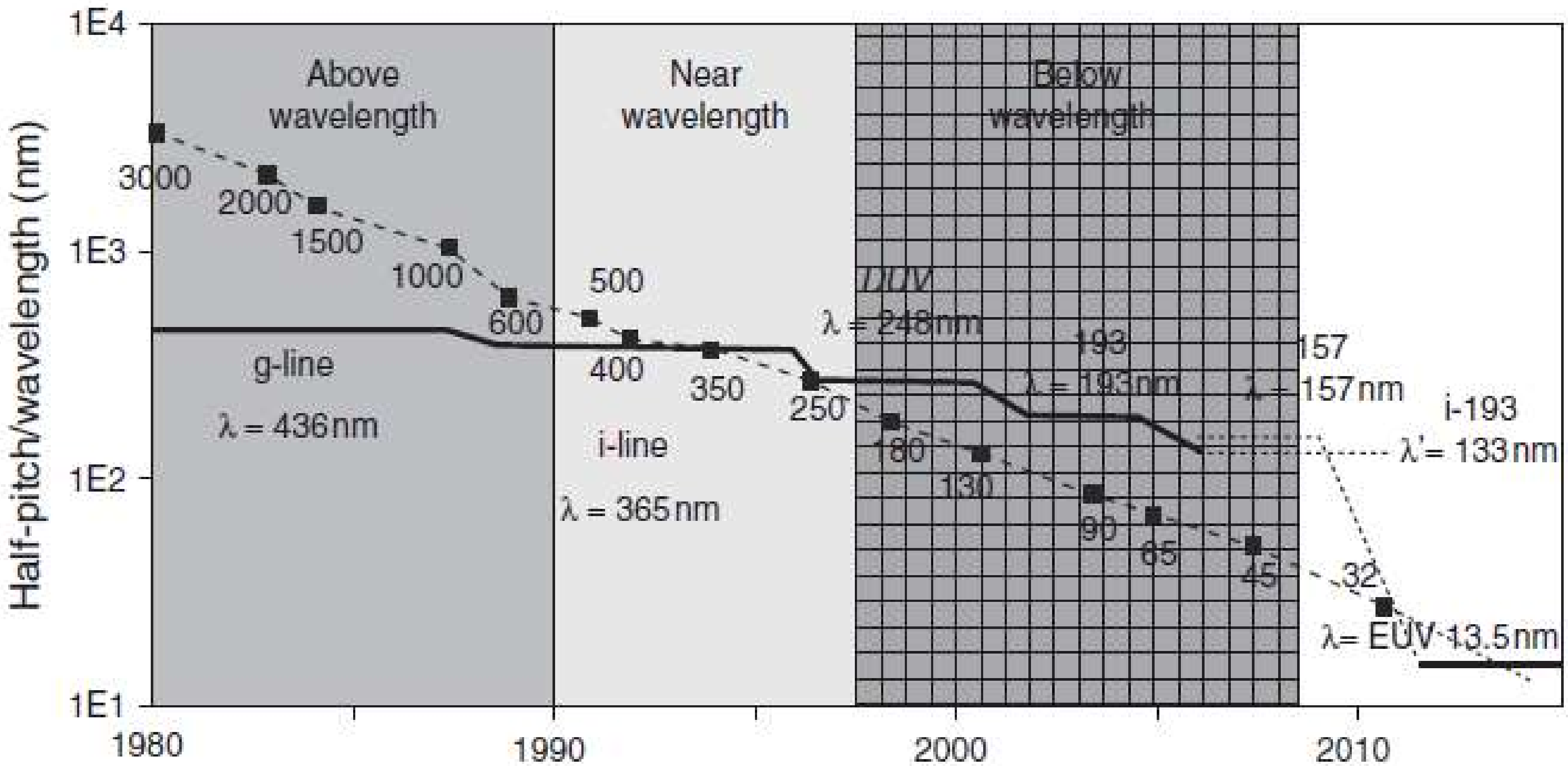
Although usually hundreds of microprocessors are built on a single wafer, this picture story will only focus on a small piece of a microprocessor from now on –on a transistor or parts thereof.

Washing off of Photo Resist



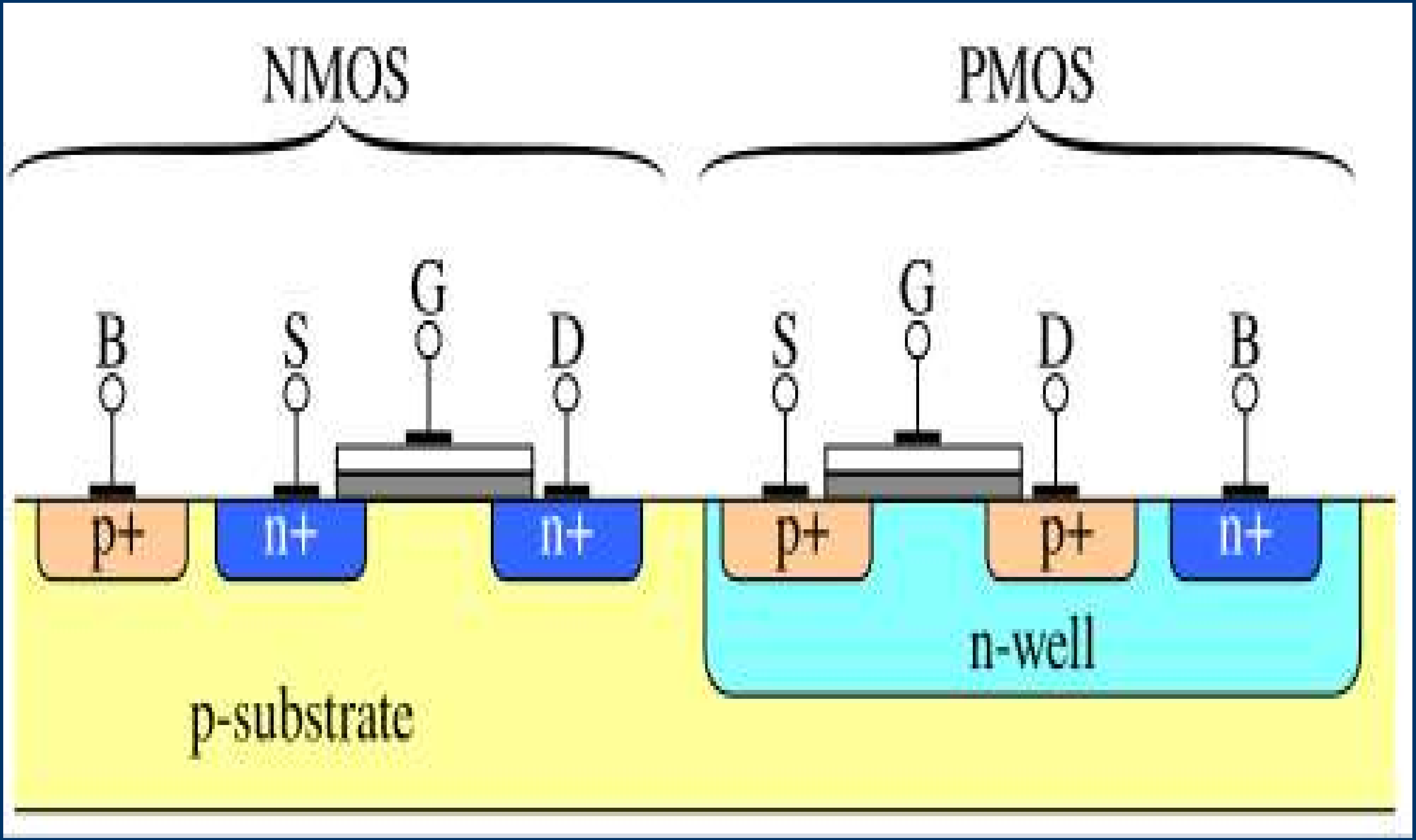
The gooey photo resist is completely dissolved by a solvent. This reveals a pattern of photo resist made by the mask.

Lithography Challenges

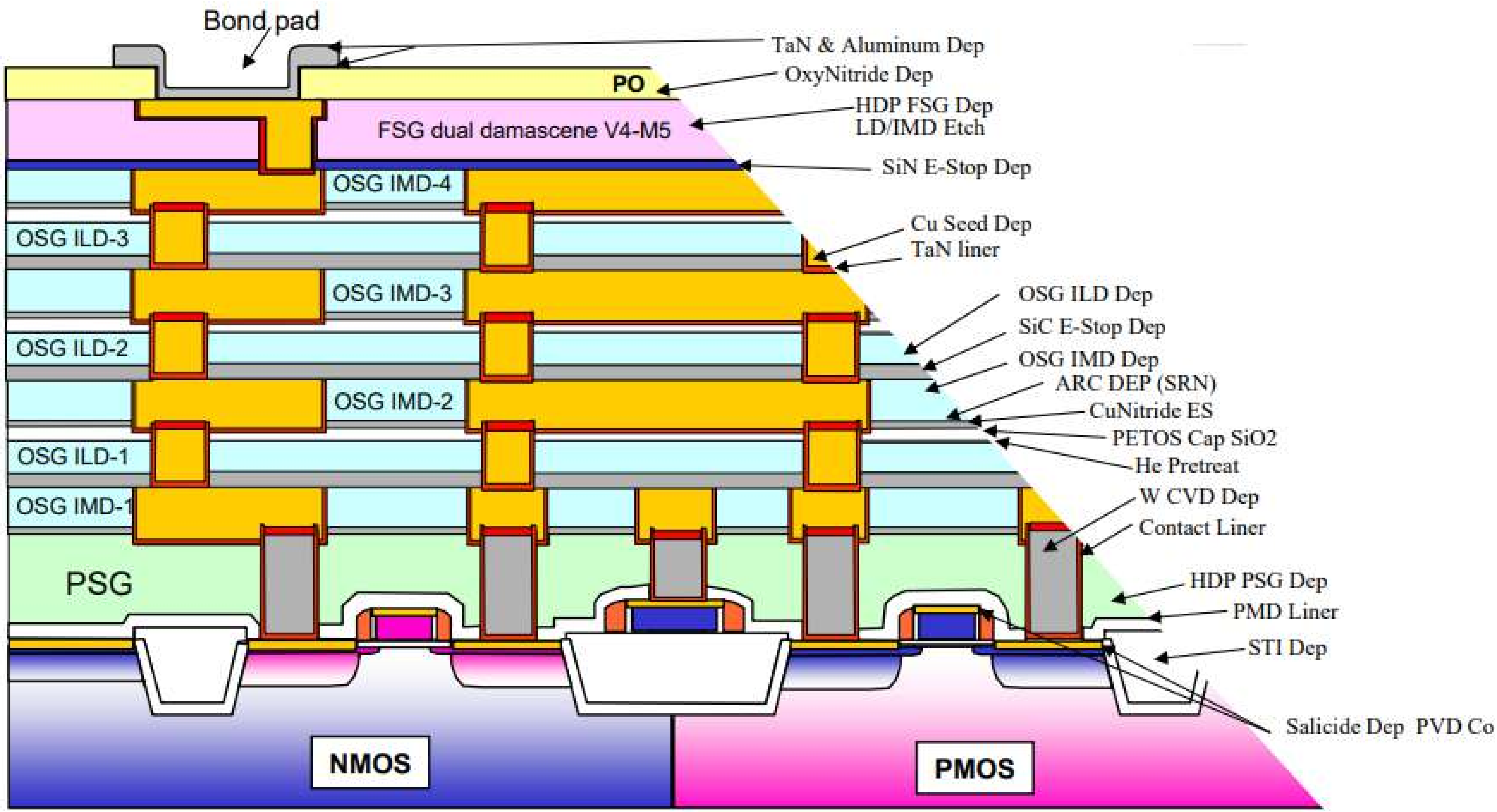


Kundu *2010 McGraw Hill*

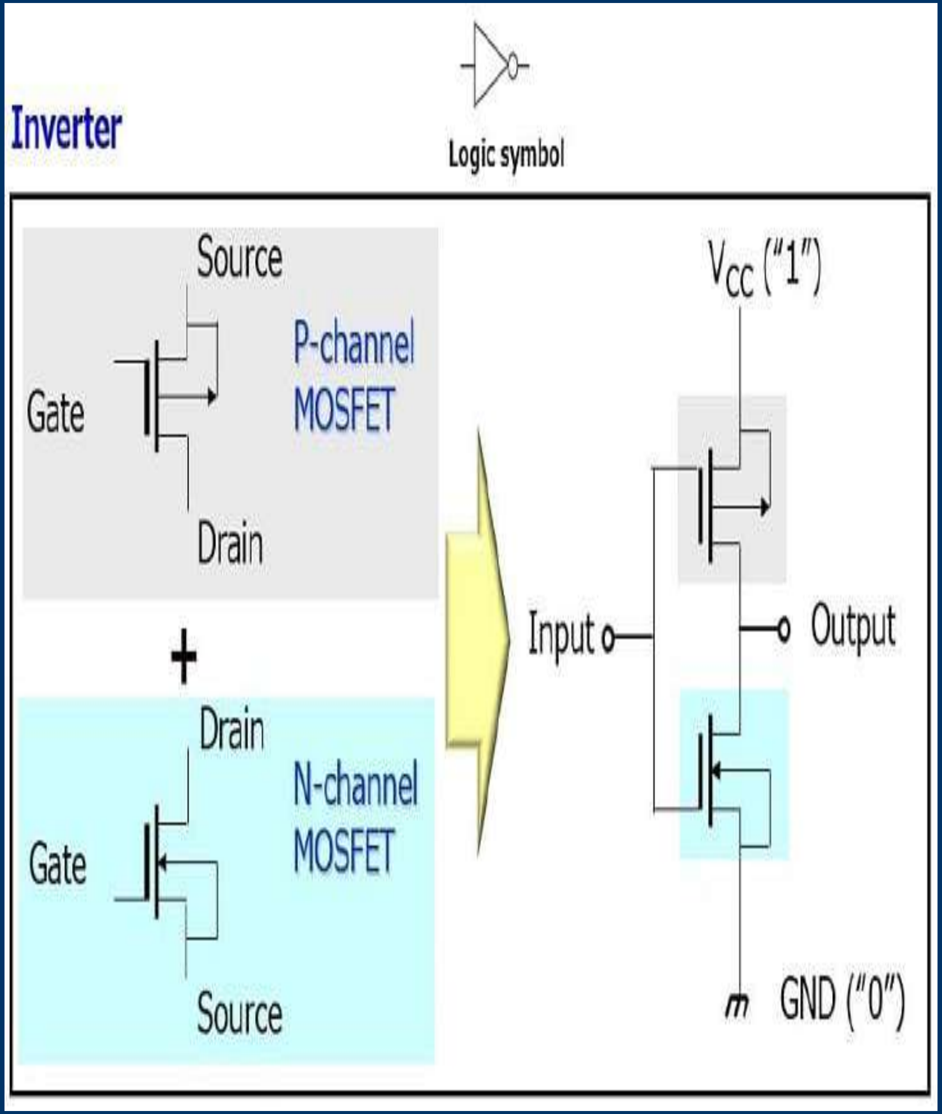
Silicon Complimentary Metal-oxide –silicon (CMOS) Transistor



Interconnect Integration



Basic CMOS Inverter Circuit

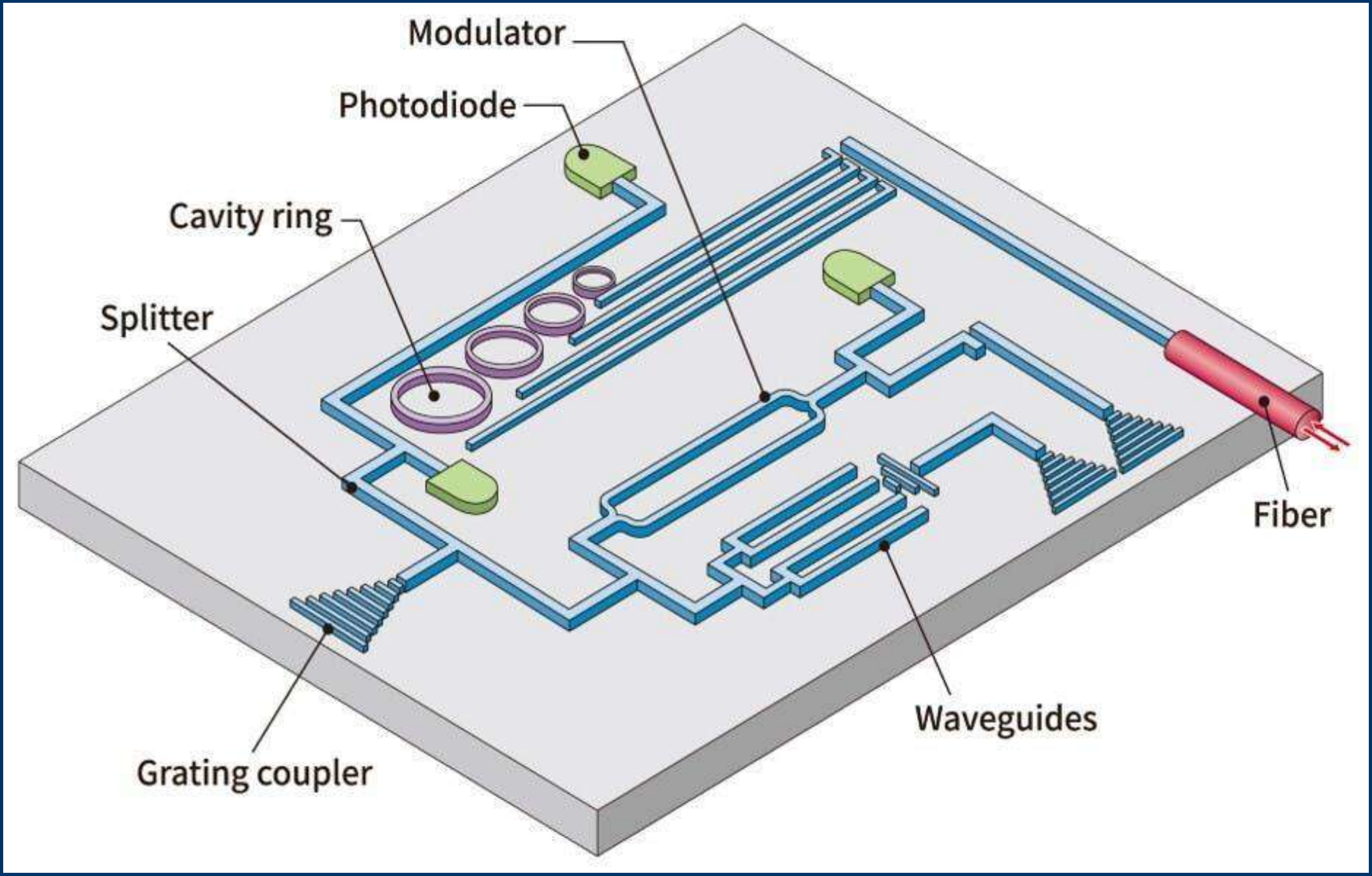


- Current in Both n-MOSFET and p-MOSFET should be equal.
- Importance of the Mobility

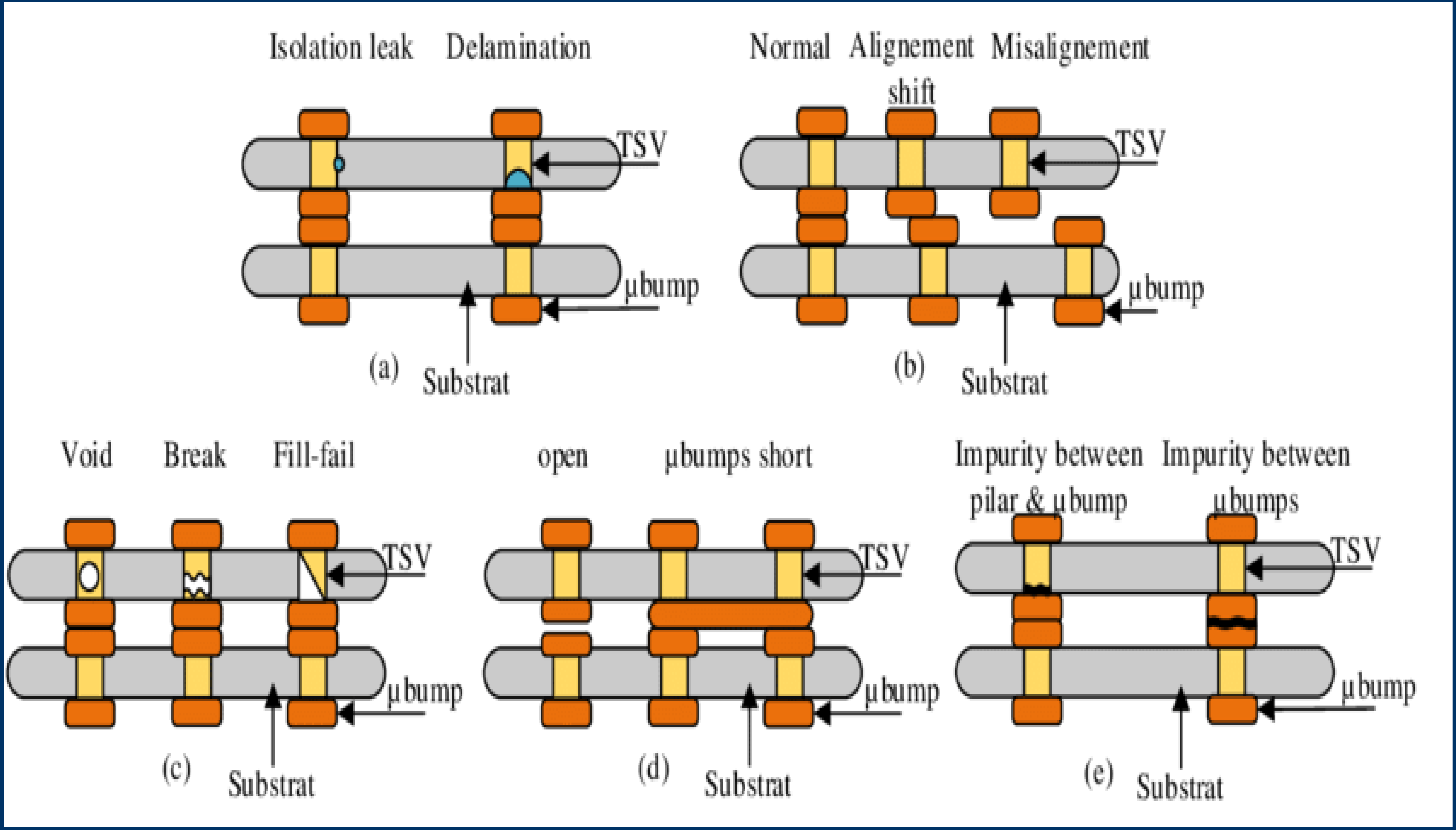
$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} = \frac{\mu_n}{\mu_p}, \qquad (3.5\text{-}38)$$

Since $\mu_n/\mu_p \simeq 2.5$, a minimum-area CMOS inverter will have $(W/L)_n \simeq 1$ and $(W/L)_p \simeq 2.5$.

Silicon Photonics Based on CMOS Manufacturing



Defects: Microscopic and Macroscopic Process Induced Defects



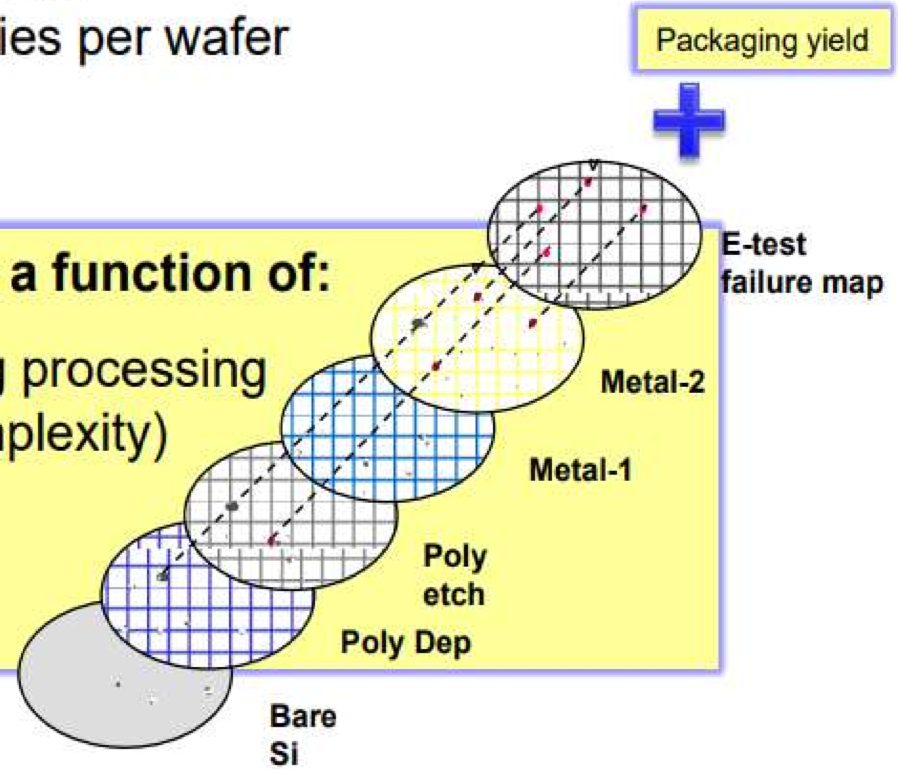
What is yield?

$$Y = \frac{N_{\text{good}}}{N_{\text{total}}}$$

N_{total} = number of dies per wafer
N_{good} = number of working dies per wafer

The yield of semiconductor fabrication is a function of:

- The defects per unit area introduced during processing (number of process steps and process complexity)
- Statistical defect distribution
- Die area (device size)



Calculate Y_R with Yield Y_i=0.99 and 50 steps.

$$Y_R = Y_{Si} * Y_{poly} * Y_{m0} * Y_{m1} * Y_{m2} * \ldots$$

Process Control

http://electroiq.com/blog/2016/02/yield-and-cost-challenges-at-16nm-and-beyond

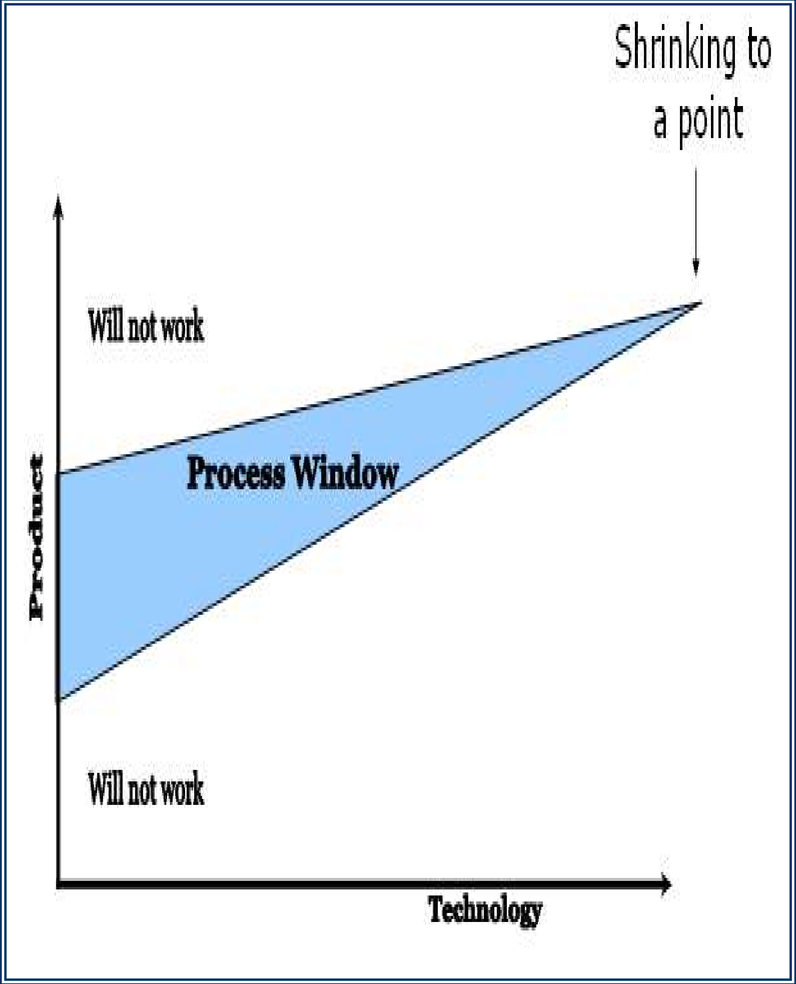
The Ten Fundamental Truths of Process Control

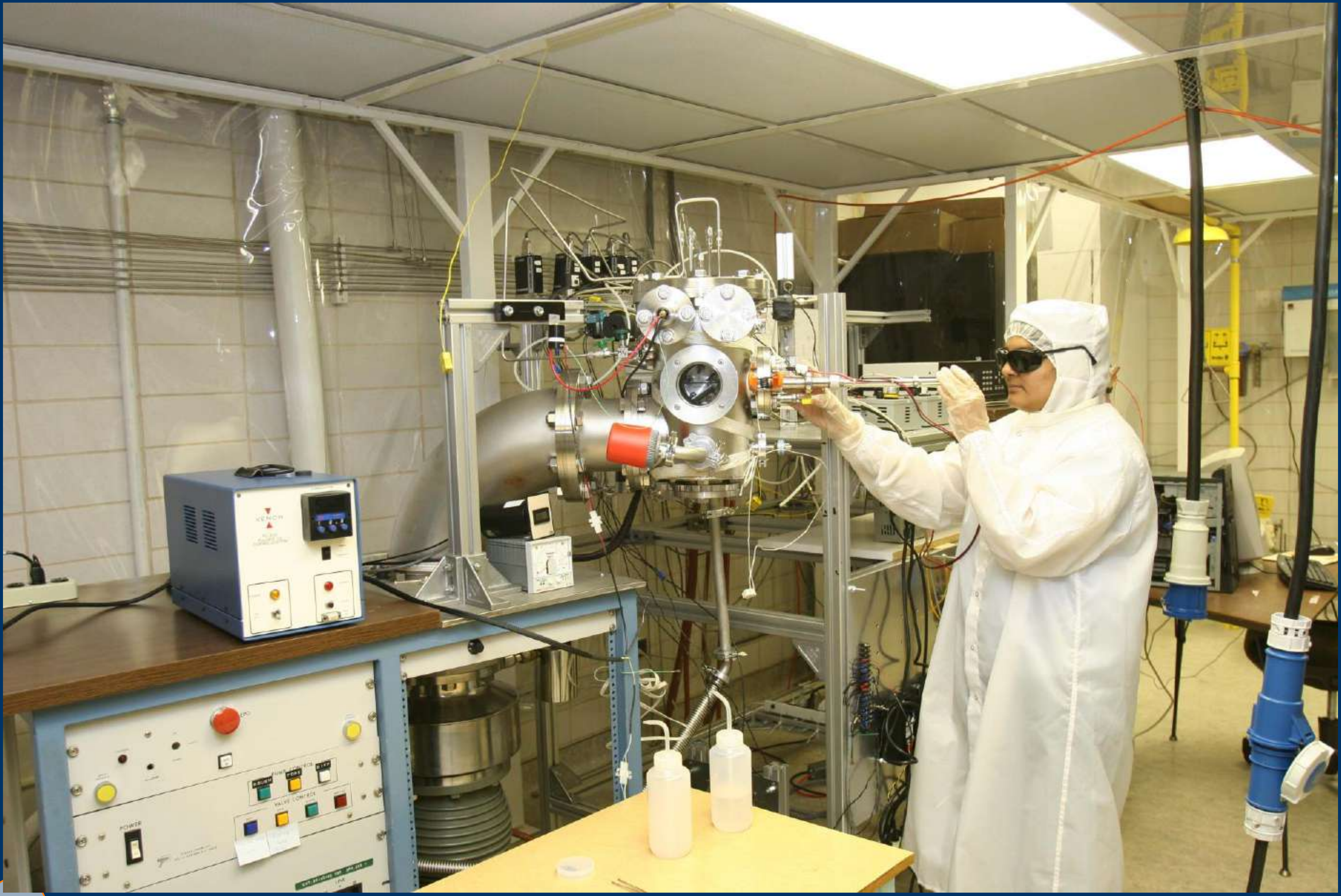
1. You can't fix what you can't find. You can't control what you can't measure.
2. It is always more cost-effective to over-inspect than to under-inspect.
3. The most expensive defect is the one that wasn't detected inline.
4. Always quantify your lots at risk when making changes to your process control strategy.
5. Variability is the enemy of a well-controlled process.
6. Time is the enemy of profitability.
7. Improving yield also improves device reliability.
8. Process control requirements increase with each design rule.
9. High-stakes problems require a layered process control strategy.
10. Adding process control *reduces* production costs and cycle time.

FIGURE 1. The ten fundamental truths of process control for the semiconductor IC industry.

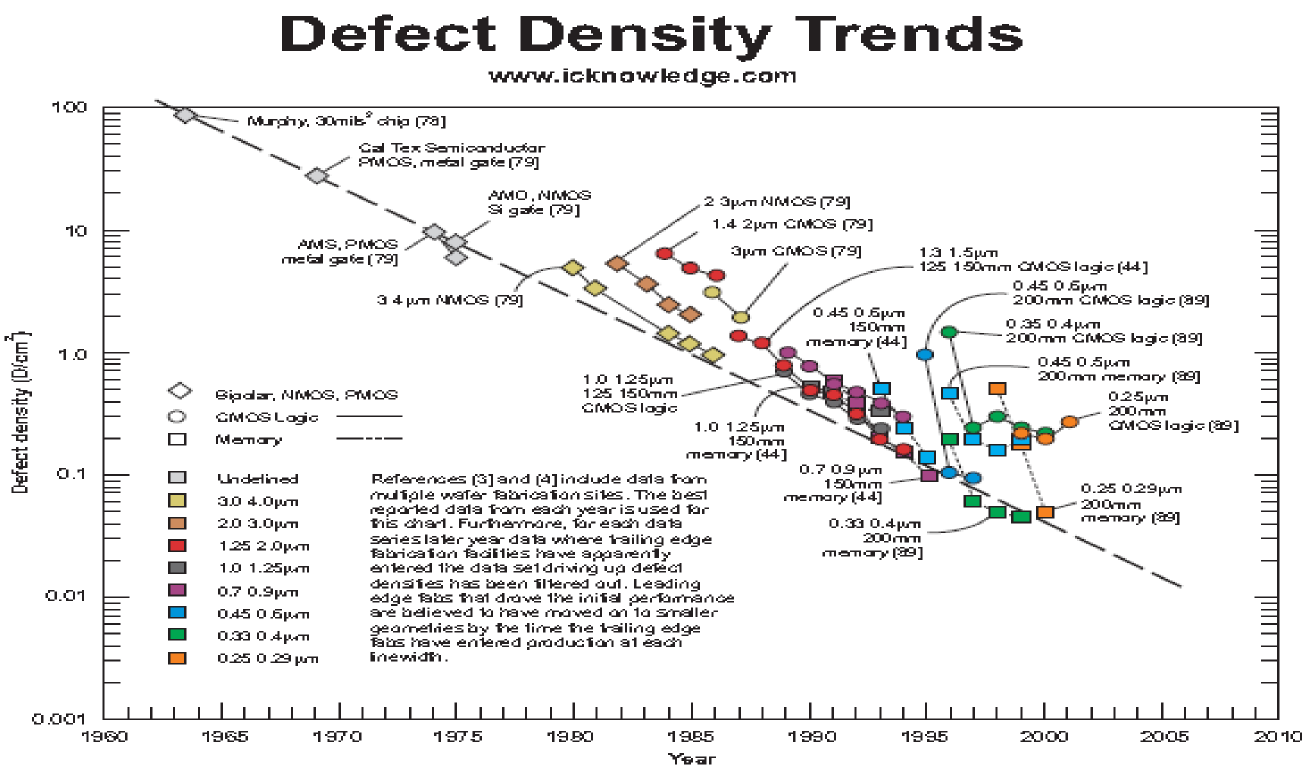
Challenges : 1-D & 2-D Structures

- Process Window Shrinking to a Point "Need for a High Level of Process Control which is Beyond Current Techniques" R. Singh et. al. Semiconductor Fabtech, 20th Edition, pp. 104, 2006,





Reduced Defect Density & Lower Process Variability



[1] Philip R. Thomas, "Competitiveness Through Total Cycle Time," McGraw Hill (1990).
[2] James A. Cunningham, "The Remarkable Trend in Defect Densities and Chip Yields," Semicon. International, p.86, June (1992).
[3] Robert C. Leachman ed., "Competitive Semiconductor Manufacturing Survey. Third Report of the Main Phase," UC Berkeley (1996).
[4] Robert C. Leachman, "Competitive Semiconductor Manufacturing. Final Report from benchmarking Eight-inch, sub-350nm Wafer Fabrication Lines," UC Berkely (2002).

14073a

IC KNOWLEDGE.COM

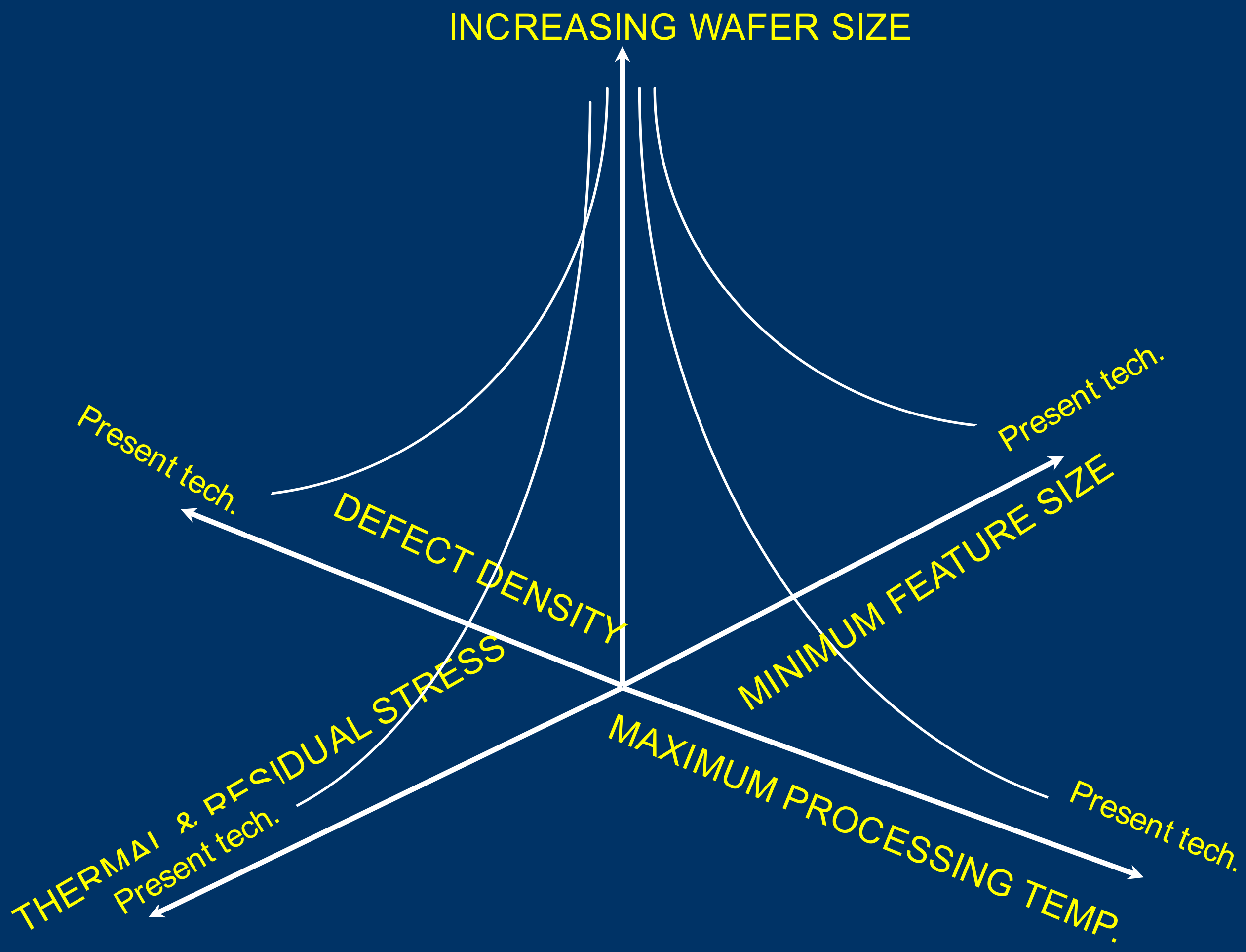
Low Cost of Ownership (COO)

- COO = (CF+CV+CY)/(TPT x Y x U)
- CF = Fixed cost
- CV = Variable cost
- CY = Cost due to yield loss
- TPT = Throughput
- Y = Composite Yield =exp(-DA) where A is device area and D is the defect density
- U = Utilization

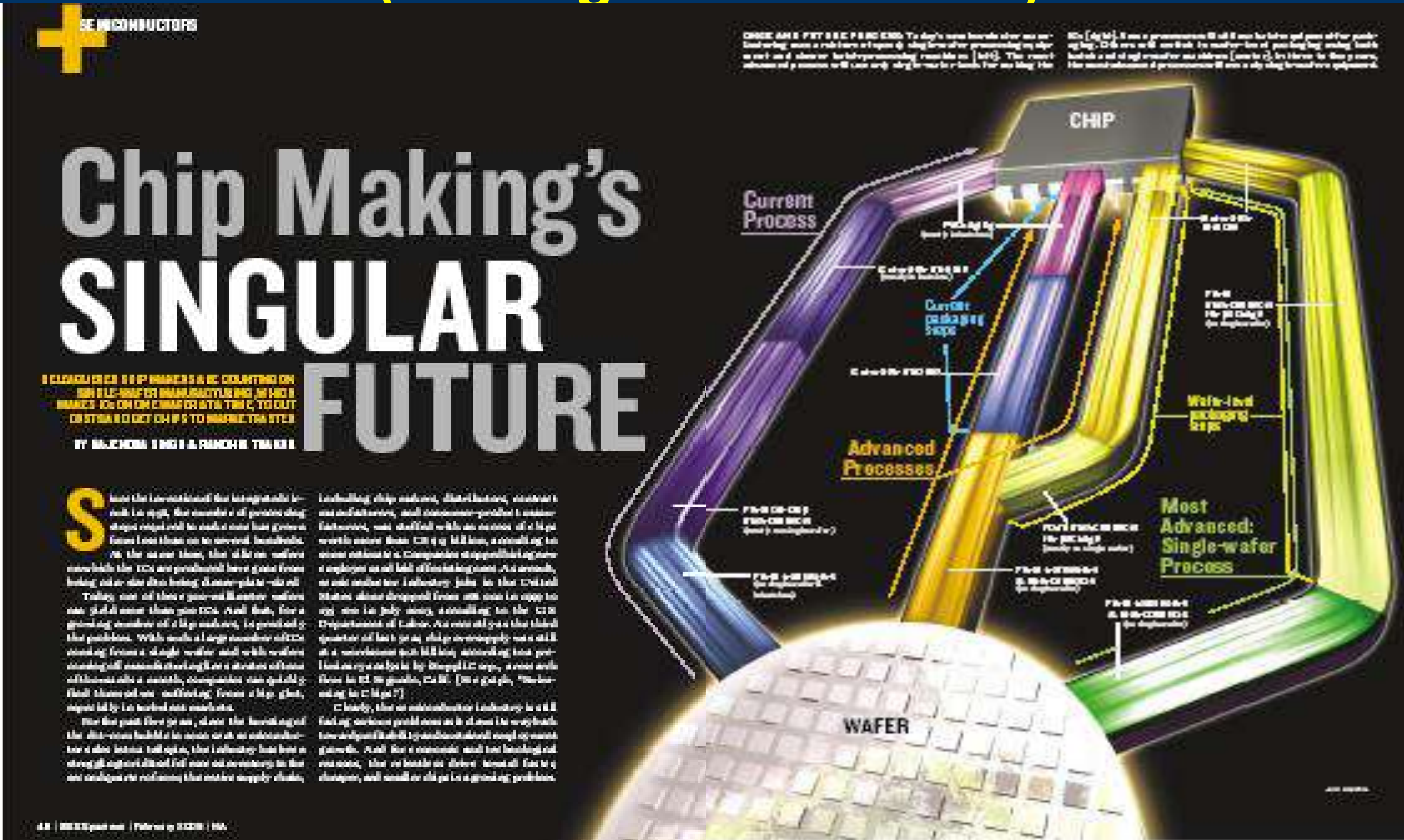
KEY FUNDAMENTALS OF SEMICONDUCTOR MANUFACTURING

Semiconductor Manufacturing in 21st Century: Nanostructure Homogeneity for highest Performance, Reliability and Yield

R. Singh, et. al. Semiconductor Fabtech, 9th Edition, p. 223, 1999



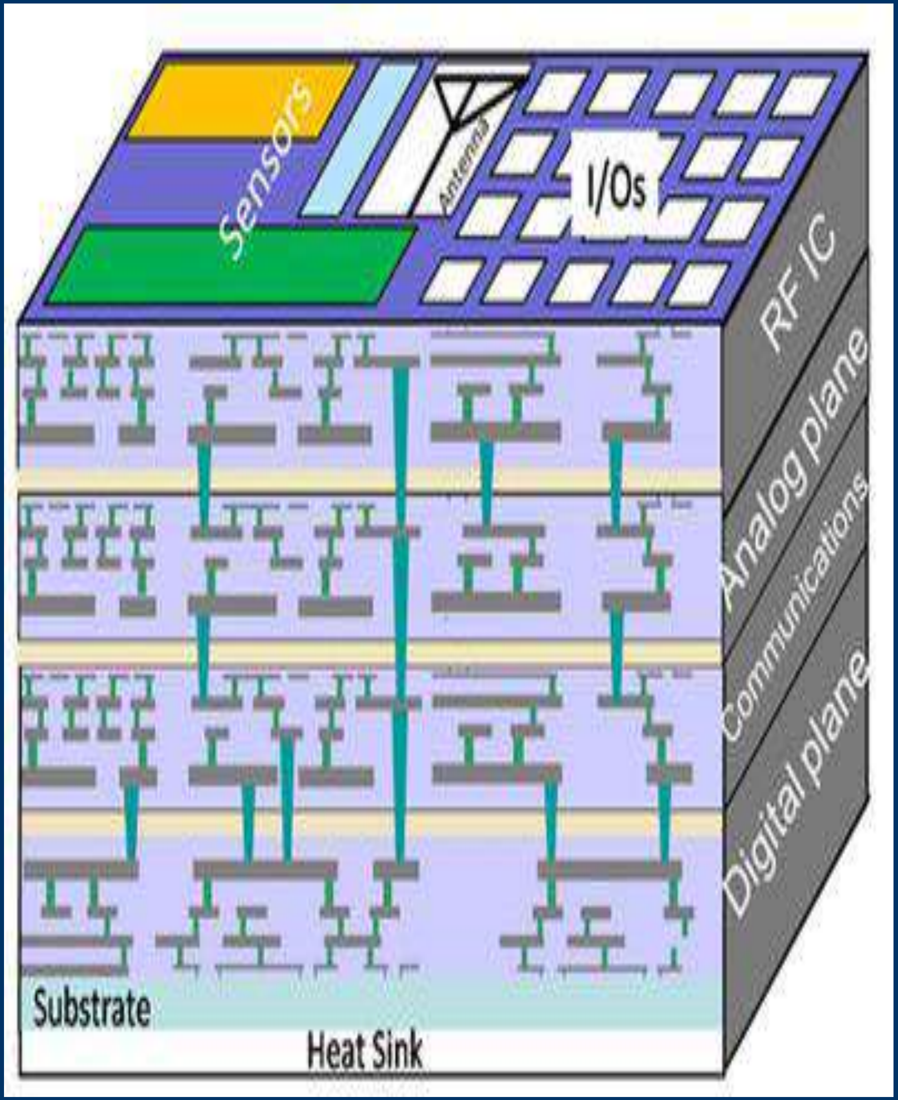
IEEE Spectrum February 2005
(R. Singh & R. Thakur)



Packaging: 3-D INTEGRATION

- Three-Dimensional devices, such as the 3D-Integrated Circuit (3D-IC)

- Advantages of 3D systems
  - Form factor
  - Integration
  - Performance



How Small ?

THEORETICAL WORK

- Length: 4.05 x 10^{-35} m (Planck Distance)
- Time: 1.35 x 10^{-43} s (Planck Time)
- Planck scale number set the ultimate limits on the performance of computers
- With black holes as the potential material, the switching time of the order of 10^{-43} s is possible
- Source: N. Gershenfeld "The Physics of Information Technology", Cambridge Press

REAL WORLD

- Current: 1.0 x 10^{-15}A
- Time: 1.0 x10^{-15} s (~ 1.0 x 10^{-18} s)
- One Atom (~0.1 nm diameter)

ULTIMATE DISRUPTIVE TECHNOLOGY THAT WILL PROVIDE LEAFFROG CAPABILITES TO BHARAT

Examples of Disruptive Technologies

- 1982: Discovery of Scanning Tunneling Microscope followed by discovery of AFM in 1985
- 1986: Noble Prize in Physics and the rest is history
- 1987: Dave Payne at University of Southampton develops erbium-doped fiber amplifier operating at 1.55 micrometers
- 1990: Erbium-doped Fiber Amplifier (EDFA), broadband amplification in practice and the rest is history

Life: Perspective

Matter	Dimension (nm)
Atom	0.1
DNA Width	2
Protein	5-50
Virus	75-100
Materials internalized by cells	< 100
Bacteria	1,000-10,000
White Blood Cell	10,000

Semiconductor Fabtech, 20th Edition, p. 104, 2006

IG

Dominance of silicon CMOS based semiconductor manufacturing beyond international technology roadmap and for many more decades to come

R. Singh, **P. Chandran**, **M. Grujicic**, **K.F.Poole**, **U. Vingnani**, **S.R. Ganapathi**, **A. Swaminathan**, **P. Jagannathan**, & **H. Iyer**, Clemson University, South Carolina, USA

Fundamentals of Disruptive Nanoelectronics (2002)

JOURNAL OF NANOSCIENCE AND NANOTECHNOLOGY

Fundamental Device Design Considerations in the Development of Disruptive Nanoelectronics

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In the last quarter of a century silicon-based integrated circuits (ICs) have played a major role in the growth of the economy throughout the world. A number of new technologies, such as quantum computing, molecular computing, DNA molecules for computing, etc., are currently being explored to create a product to replace semiconductor transistor technology. We have examined all of the currently explored options and found that none of these options are suitable as silicon IC's replacements. In this paper we provide fundamental device criteria that must be satisfied for the successful operation of a manufacturable, not yet invented, device. The two fundamental limits are the removal of heat and reliability. The switching speed of any practical man-made computing device will be in the range of 10^{-15} to 10^{-3} s. Heisenberg's uncertainty principle and the computer architecture set the heat generation limit. The thermal conductivity of the materials used in the fabrication of a nanodimensional device sets the heat removal limit. In current electronic products, redundancy plays a significant part in improving the reliability of parts with macroscopic defects. In the future, microscopic and even nanoscopic defects will play a critical role in the reliability of disruptive nano-electronics. The lattice vibrations will set the intrinsic reliability of future computing systems. The two critical limits discussed in this paper provide criteria for the selection of materials used in the

Open Question: Can we have Disruptive Nanoelectronics?

- The Speed of transistor and heat generated are governed by Heisenberg uncertainty Principle
- Ultimate lithographic limit is governed by Heisenberg uncertainty Principle
- The ultimate reliability of computing system is governed by Lattice Vibrations
- Minimum value of power supply is 3KT/q (~78 mV@ 300K)
- Based on quantum considerations, one needs 40 particles for three terminal logic device operating at room temperature
- Due to gain issue two terminal devices are not suitable for logic applications
- With current low-cost materials we can not have packages that can handle more than 150 Watt
- Thermal conductivity of semiconductor plays a very important role

HEAT DISSIPATION

- Slow speed computers pose no fundamental limit
- Human brain with neural density ~ 10^6 per cm^2 , speed ~10^{-3} s handles a power density of 250 nW/cm^2 very well
- The design of faster and denser computers is dictated by power handling capability
- **Cooling Chips Still A Top Challenge (May 22, 2025)**
- https://semiengineering.com/cooling-chips-still-a-top-challenge/

- **Liquid Cooling Gains Traction In Data Centers (January 15, 2025)**
- https://semiengineering.com/liquid-cooling-gains-traction-in-data-centers/

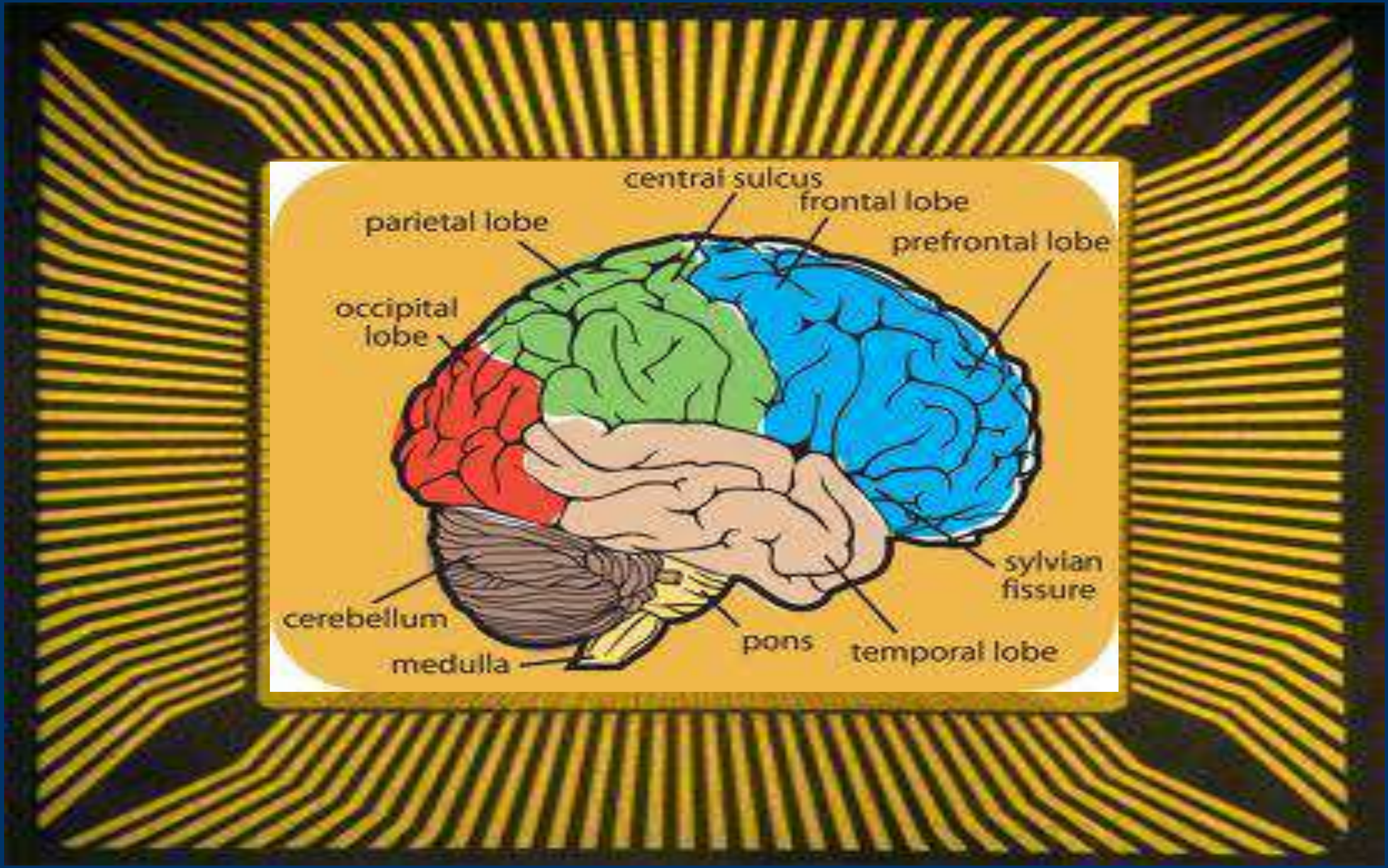
Requirements of Nano-diamond Technology

PROCESS	
Substrate size	12 inch diameter
Feature size	2-5 nm
Thermal conductivity	>Silicon
Integration density	$>10^{11}$ elements/cm^2
PERFORMANCE	
Operating Temperature	Room temperature
Power consumption	$<10^{-3}$ fJ/switch
Gain	>10
Speed	<100fS
Reliability	<0.1FIT (one in billion)

Mimicking Human Brain

- Replacement of Von Neumann Architecture
- Fuzzy Logic
- Artificial Intelligence
- Expert Systems
- Artificial Neural Network (ANN)
- ANN Software Approach (Less Efficient)
- ANN Hardware Approach (More Efficient)

Building a Brain On a Silicon Chip: Require New Architecture which no one has discovered so far



Quantum Computers: Current Know how

- QUANTUM COMPUTERS: Since error correction is part of quantum computing, more heat will be generated in a quantum computer than a Von-Neumann architecture based computer operating at the same speed.
- Assuming that researchers can lengthen the decoherence interval to 10 microseconds; quantum computer chips will still consume more than 100 megawatts.
- The facts presented here indicate that quantum computers if ever realized will involve massive size and extra ordinary cooling techniques (e.g. cooling of a nuclear reactor) needs to be implemented.

Ultimate Computer ?

- Nano-diamond based C-MIS FET deposited on Silicon Substrate
- Nano-diamond is the only material that can meet all other fundamental requirements and provide complimentary FET(μ_n= 2200 cm^2/V-s and μ_p= 1600 cm^2/V-s) in line with the legacy of silicon CMOS
- Best for every applications : high temperature, high performance computing , high frequency, and high and low-power Nanoelectronics (including quantum computers and AI Computing)

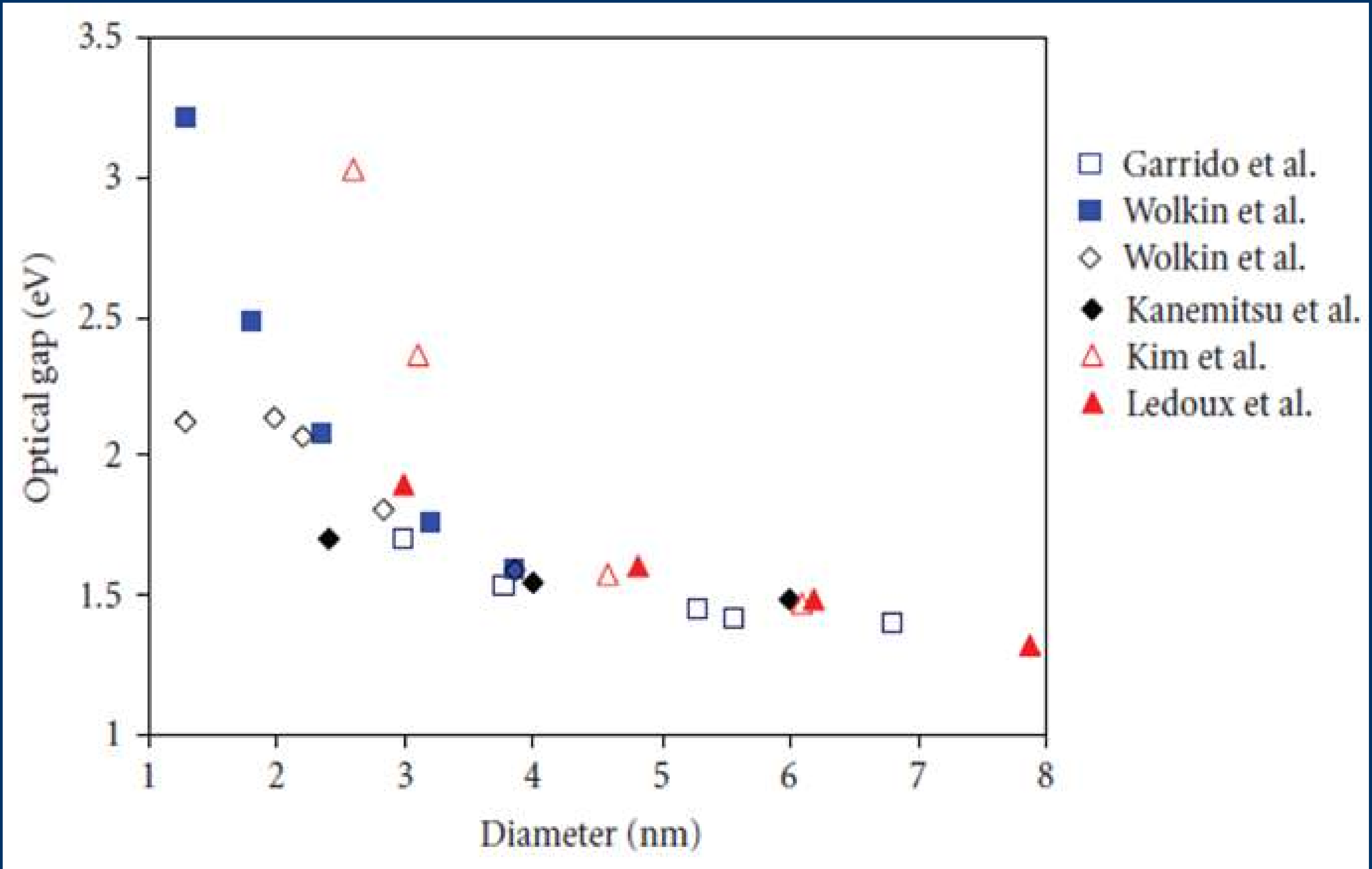
Why There is no Diamond Based Electronics?

- Due to the strongest bond (C-C) it is difficult to grow single crystals (nature needs high pressure, and high temperature to grow crystals with defects)
- Due to wide band gap, need of new invention of doping and making ohmic contacts

Nanostructure Silicon Provides the Most Difficult & Most Rewarding Opportunity in Solid State

R. Singh, F. Alapatt and A. Lakhtakia, IEEE JEDS, vol. 1(6) pp. 129-144 , 2013

https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=6589128



CONCLUSIONS

- The World of Silicon: It's Dog-Eat-Dog (IEEE Spectrum, p. 30, Sept. 1988)
- VERY COMPETITIVE INDUSTRY
- There is nothing more complex than manufacturing 1.4-2 nm chips
- Cost, Cost, Cost
- $,$, $
- Bharat is providing new opportunities to all new Engineers in energy, computing, communication, manufacturing etc. driven by semiconductor industry
- Taking Bharat to next level of disruptive semiconductor technology needs your dedicated commitment to succeed
- Do not be afraid of failure, but thinking and acting on big and difficult problem is very rewarding.

We are Sitting at the Tip of the Iceberg

https://www.google.com/search?q=We+are+Sitting+at+the+Tip+of+the+Iceberg&client=firefox-a&hs=i2S&rls=org.mozilla:en

Thank You.

QUESTIONS